

Highlights

REVEX: A Unified Framework for Removal-Based Explainable Artificial Intelligence in Video

F. Xavier Gaya-Morey, Jose M. Buades-Rubio, I. Scott MacKenzie, Cristina Manresa-Yee

- Developed REVEX, a framework for removal-based video explanations in XAI
- Stepwise analysis reveals strengths and weaknesses of removal-based methods
- Adapted six XAI methods to video, integrating temporal information
- Video LIME excelled in deletion/insertion metrics; Video RISE in average drop metric
- Evaluated XAI metric limitations and their suitability for various applications

REVEEX: A Unified Framework for Removal-Based Explainable Artificial Intelligence in Video

F. Xavier Gaya-Morey^{a,b,c,*}, Jose M. Buades-Rubio^{a,b,c}, I. Scott MacKenzie^d, Cristina Manresa-Yee^{a,b,c}

^a*Computer Graphics and Vision and AI Group (UGIVIA), Universitat de les Illes Balears, Carretera de Valldemossa, km 7.5, Palma, Illes Balears, 07122, Spain*

^b*Research Institute of Health Sciences (IUNICS), Universitat de les Illes Balears, Carretera de Valldemossa, km 7.5, Palma, Illes Balears, 07122, Spain*

^c*Department of Mathematics and Computer Science, Universitat de les Illes Balears, Carretera de Valldemossa, km 7.5, Palma, Illes Balears, 07122, Spain*

^d*Department of Electrical Engineering and Computer Science, York University, 4700 Keele St, Toronto, M3J 1P3, Canada*

Abstract

We developed REVEEX, a removal-based video explanations framework. This work extends fine-grained explanation frameworks for computer vision data and adapts six existing techniques to video by adding temporal information and local explanations. The adapted methods were evaluated across networks, datasets, image classes, and evaluation metrics. By decomposing explanation into steps, strengths and weaknesses were revealed in the studied methods, for example, on pixel clustering and perturbations in the input. Video LIME outperformed other methods with deletion values up to 31% lower and insertion up to 30% higher, depending on method and network. Video RISE achieved superior performance in the average drop metric, with values 10% lower. In contrast, localization-based metrics revealed low performance across all methods, with significant variation depending on network. Pointing game accuracy reached 53%, and IoU-based metrics remained below 20%. Drawing on the findings across XAI methods, we further examine the limitations of the employed XAI evaluation metrics and highlight their

*Corresponding author.

Email addresses: `francesc-xavier.gaya@uib.es` (F. Xavier Gaya-Morey), `josemaria.buades@uib.es` (Jose M. Buades-Rubio), `mack@yorku.ca` (I. Scott MacKenzie), `cristina.manresa@uib.es` (Cristina Manresa-Yee)

suitability in different applications.

Keywords: explainable artificial intelligence, action recognition, deep learning, computer vision

1. Introduction

As deep learning (DL) gains importance across many domains, the need for explainable artificial intelligence (XAI) methods emerges. XAI endeavors to both enhance our comprehension of “black box” models and to foster user trust in their utilization, all while upholding robust learning performance [1].

Recent studies in computer vision focus on explaining DL models. Nevertheless, despite progress with image models (e.g., [2, 3, 4]), explainability in video remains relatively under-explored. The scarcity of attention is likely due to the added temporal dimension in video data. This escalates the complexity of both the data and the models to analyze them.

Although some studies explore video explanations, attempts often lean toward convolutional architectures, typically employing back-propagation (e.g., [5, 6, 7]). However, our research is directed at model-agnostic explanations that extend beyond architecture-specific solutions. Although model-agnostic explanation techniques exist for images, for example, LIME [8], RISE [9], or SHAP [10], their application to the video domain presents challenges. With video, the temporal dimension is added. This escalates computational demands, necessitating careful adaptation. To address this, we first identify common elements in the explanation process. This enables more controlled adjustments for video data while regulating their influence on the explanations. The primary contributions of this work are threefold:

- A refined, fine-grained framework for explaining model predictions tailored to computer vision, and unifying common explanation steps in existing methods.
- An adaptation of six established explanation methods to the video domain, modifying each to accommodate the temporal dimension and local explanations.
- A critique of the explanations generated from each method, including diverse models and datasets, to robustly assess their performance.

This paper is organized as follows. First, existing video and removal-based explanation methods are surveyed. Subsequent sections explore our REVEX framework for video explanations, breaking the steps into well-defined stages. We then extend six explanation methods to the video domain, conducting experiments with various networks and datasets. Subsequently, we critique the explanations. Finally, we offer insights gleaned from our research. Code to reproduce the experiments and analyses is available at https://github.com/Xavi3398/revex_framework.

2. Related Work

2.1. Video Explanation Methods

Despite extensive research on image-based explanation methods, video explanations are comparatively under-explored. Table 1 summarizes studies on video input.

Table 1: Research on Explanation Methods Using Video Data

Ref.	Year	1st Author	Explanation Method
[11]	2024	Gaya-Morey	LIME separated into spatial and temporal explanations
[12]	2023	Uchiyama	Adaptive occlusion sensitivity analysis (AOSA)
[5]	2022	Hartley	Superpixels weighted by average gradients for video (SWAG-V)
[13]	2021	Li	Spatio-temporal extremal perturbation (STEP)
[7]	2020	Hiley	Selective relevance
[14]	2019	Stergiou	Class feature pyramids
[6]	2019	Stergiou	Saliency tubes
[15]	2019	Hiley	Discriminative relevance
[16]	2018	Bargal	Contrastive excitation backpropagation for RNNs (cEB-R)
[17]	2018	Anders	Deep Taylor decomposition [18]
[19]	2017	Srinivasan	Layer-wise relevance propagation (LRP) [3]
[20]	2015	Gan	Class saliency maps [2]

Gaya-Morey et al. [11] applied LIME (local interpretable model-agnostic explanations) to video data by employing grid-based, separated spatial and temporal perturbations. By separately analyzing the spatial and temporal dimensions, the individual contributions to predictions were revealed. This reduced the time for explanations by merging the separate explanations, rather than computing them across all dimensions simultaneously. However, results degrade with increased temporal variation.

Uchiyama et al. [12] extended OSM (occlusion sensitivity method) [21] from images to videos. In the image approach, a gray patch is manipulated through an image to gauge its impact on predictions. For video, they introduced optical flow to extend the two-dimensional patch across the temporal dimension, aiming to trace object trajectories through video sequences.

Hartley et al. [22] proposed SWAG (superpixels weighted by average gradients), which extends SLIC (simple linear iterative clustering) [23] to compute superpixels using gradients. Relevance was assessed using guided back-propagation [24]. Their SWAG-V extension adds the temporal dimension (i.e., video) during superpixel assignment [5].

Li et al. [13] proposed STEP (spatio-temporal extremal perturbation), an extension of extremal perturbation (EP) [25]. EP optimizes explanation masks, subject to constraints. The extension accommodates the temporal dimension in video data.

Hiley et al. [15] explored video explanations using deep Taylor decompositions [18]. They disentangled spatial and temporal relevance by explaining individual frames expanded to input dimensions across video frames, thus simplifying temporal information. They identified weaknesses in this approach in a later study [7] and proposed selective relevance to decompose a 3D explanation into spatial and temporal components. This was done via derivative-based filtering to discard regions with near-constant relevance over time.

Stergiou et al. introduced two methods for explaining 3D convolutional neural networks (CNNs). First [6], they presented saliency tubes, utilizing prediction outputs and the final activation layer to compute saliency maps rescaled to match input dimensions. Second [14], they introduced class feature pyramids, which use back-propagation to traverse the network, identifying significant kernels at different depths. The resulting saliency maps yielded higher resolution than the saliency tubes approach.

Bargal et al. [16] used recurrent neural networks (RNNs) for video action recognition and captioning with a contrastive excitation back-propagation formulation. This was the first instance of top-down saliency formulation within deep recurrent models, facilitating space-time grounding in video data.

Andersen et al. [17] delved into video prediction explanation using deep Taylor decomposition [18]. They identified pertinent pixels and frames within a video and addressed systematic imbalances, termed “border” and “look-ahead” effects, that occur in explanations.

Finally, Srinivasan et al. [19] investigated layer-wise relevance propagation

(LRP) [3] for video explanations, and Gan et al. [20] extended Simonyan et al.’s 2D saliency maps [2] to video data, yielding spatial-temporal saliency maps from 3D CNNs.

2.2. Explanations by Removal

Removal-based explanation methods, also referred to as perturbation or input-modification methods, share a common purpose: assessing the importance of input features by analyzing the change in predictions when the features are removed. This approach is model-independent since modifications are in the input space, with relevance based solely on output predictions. Covert et al. [26] analyzed 26 such methods and found three pivotal choices for a removal-based explanation paradigm: (i) feature removal, (ii) model behavior, and (iii) feature assessment. While some methods are well-established for image data, their adaptation to video entails additional considerations. Given the high volume of input features, particularly in video, it is common to use sets of pixels as the basis for explanation. Additionally, the unique characteristics of visual data necessitate adjustments, such as replacing pixels to simulate removal. These adaptations introduce multifaceted choices when designing an explanation method. Next, we examine six existing removal-based explanation methods that are adapted for the video domain: LIME, SHAP, RISE, LOCO, univariate predictors, and occlusion sensitivity.

LIME [8], or local interpretable model-agnostic explanations, was intended to elucidate predictions of diverse classifiers, with an application to images. LIME adopts superpixels as regions for segmenting images, although it does not prescribe a specific technique. It functions by randomly occluding approximately half of the chosen regions, observing variations in class confidence, and subsequently training an interpretable surrogate model, such as linear models or decision trees, to compute the relevance of each region for different target classes. Various visualization strategies are employed, from displaying only the most significant regions to using distinct colors for positive and negative relevance.

SHAP (shapley additive explanations) [10] introduces Shapley values, a unified measure of feature relevance. The values express the expected change in model prediction when conditioning on a specific feature. Although variations exist, Kernel SHAP is appealing as it yields accurate approximations with fewer passes. Kernel SHAP modifies loss functions, weighting kernels, and regularization terms from the LIME framework to derive Shapley values, ensuring local accuracy and avoiding consistency violation.

Randomized input sampling for explanation (RISE) [9], tailored for images, segments images in a small 2D grid that is scaled up through bilinear interpolation to match the image dimensions. Each grid is either occluded with a black color or left unaltered. A designated number of samples are generated, each with distinct combinations of perturbed regions. Region relevance is determined by averaging class predictions across samples with unperturbed regions. As RISE explanations are inherently positive, relevance is visualized using a heat map.

Leave-one-covariate-out (LOCO) [27] leverages feature ablation to ascertain the most influential feature. The model is trained by iteratively removing features from the training set to assess the change in prediction. LOCO offers global explanations rather than local ones and, to the best of our knowledge, has not been applied to visual data.

Univariate predictors [28] are advocated over multivariate counterparts for performance enhancements, computational efficiency, and interpretability. This method tests variable selection strategies on expected loss across the dataset. Similarly to LOCO, univariate predictors provide global rather than local explanations. They are not yet used with visual data, as far as we know.

Occlusion sensitivity [21] finds pixel relevance by sliding a gray patch across an image and assessing the change in prediction. Since each pixel is an occlusion, the pixel’s relevance correlates with the model prediction. The result is a heat map for relevance, where colder areas are regions of higher significance due to a decrease in class confidence. While this method extends to video using optical flow [12], a natural extension adds an additional dimension to the occlusion kernel. This offers advantages over optical flow, including better performance on videos containing cuts, camera movements, or objects entering or leaving the field of view.

3. The REVEX Framework: Removal-based Video Explanations

As noted above, Covert et al. [26] introduced a framework for removal-based explanations focusing on feature removal, model behavior, and feature assessment. While some of the techniques are for images, additional considerations are important in the temporal dimension of videos. Hence, prior to analyzing video-specific explanation methods, we expand upon Covert et al.’s framework by incorporating additional choices for visual data.

More specifically, we insert a segmentation choice prior to Covert et al.’s feature removal (their first choice). We then divide their feature removal

into three parts: feature selection, sample selection, and feature removal. We retain unchanged their choices for model behavior and summary techniques. Lastly, we append the visualization choice at the conclusion of the framework. A comparison of the two frameworks is presented in Figure 1, with our proposal henceforth referred to as the REmoval-based Video EXplanations (REVEX) Framework. In subsequent sections, we address each choice individually, scrutinizing advantages and limitations. We omit exploring model behavior and summary techniques, as they are addressed by Covert et al. [26].

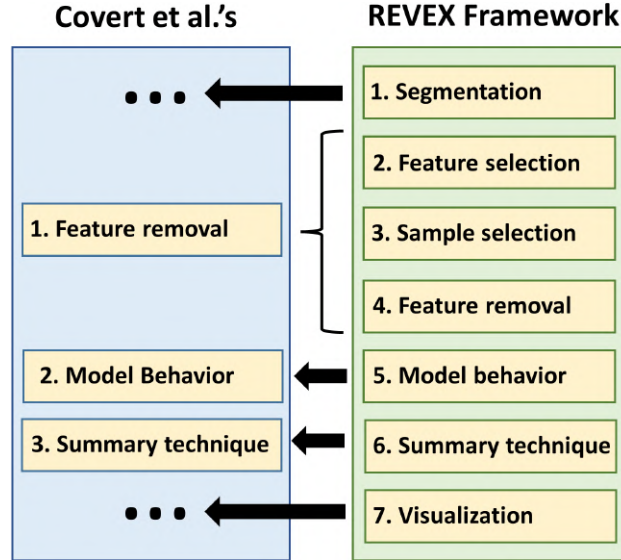


Figure 1: Covert et al.’s framework [26] vs. REVEX, our extended framework. Our proposal begins and ends with segmentation and visualization, while decomposing feature removal into three components.

3.1. Segmentation

A challenge encountered with visual explanations is the sheer volume of data. Explanation through removal operates at the level of features, which often makes operating at the pixel and frame levels infeasible. Thus, a common approach is to segment input images or videos into regions, comprising collections of pixels or pixels from diverse frames. This step significantly influences the explanations, as pixels grouped within the same region are assigned identical relevance. Figure 2 depicts strategies for video segmentation.

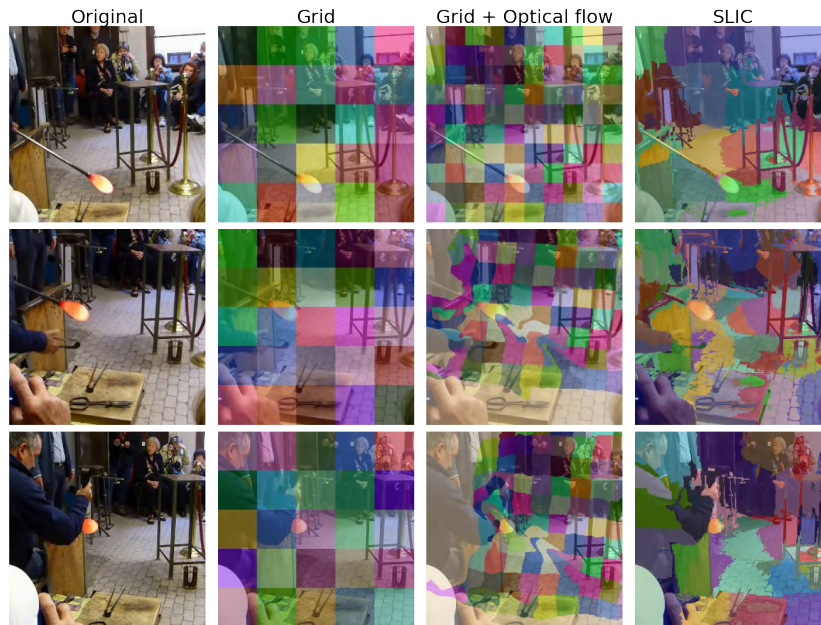


Figure 2: Video segmentation. From top to bottom, initial, middle, and final frames. From left to right, unmodified video, 3D grid, 2D grid segmentation extended with optical flow, and SLIC. Optical flow estimation was performed using PWC-Net [29].

One way to segment images or videos is through a 2D or 3D grid of cuboidal patches. While simple, it overlooks pixel color and distributes similar and proximal pixels across different regions. An example is RISE explanations [9] which extend to videos by adding the temporal dimension to the grid. See Figure 2.

Another option uses superpixels as features, wherein pixels are grouped based on similarity. Unlike grids, mentioned earlier, this method is more intricate and time-consuming, yet it gathers comparable pixels into the same regions. For videos, pixels from different frames are incorporated during superpixel computation. Figure 3 shows examples of superpixel segmentation for images, including Felzenszwalb’s efficient graph-based image segmentation [30], simple linear iterative clustering (SLIC) using K-means in a 5D color space [23], quick shift using a kernelized mean-shift approximation [31], and compact watershed utilizing marker-driven flooding to define regions with control over compactness [32]. The number of segments and their compactness (the balance between spatial and color distance) is governed by hyperparameters contingent on the method. Implementation of these tech-

niques is found in scikit-image’s segmentation package¹, although only the SLIC segmentation is for video data. The last column in Figure 2 is an example of SLIC segmentation applied to video.

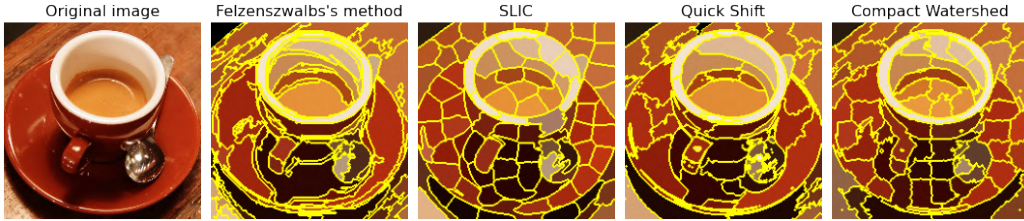


Figure 3: Example image (left) and image segmented using Felzenszwalbs’s method, SLIC, quick shift, and compact watershed. Segmented regions are outlined in yellow and filled with the average color for each region.

Finally, while many image segmentation methods extend to video, optical flow works by extending the segmentation from the first frame to the whole video by following video motion, as demonstrated by Uchiyama et al. [12] for occlusion explanations in videos. However, complications arise when segmenting objects absent in the first frame. This is evident in Figure 2 where a substantial region encompassing the individual on the left emerges towards the video’s conclusion, a result of the individual not appearing in the initial frame. Moreover, this generates segments that span the entire temporal frame, which degrades temporal explanations, as all frames are assigned the same relevance.

3.2. Feature Selection

Feature removal operates by observing the impact on the model’s predictions after removing (or occluding) features (or regions). However, certain features may exhibit a more pronounced influence when removed collectively, prompting many methods to simultaneously remove multiple features. Given that the number of features removed per sample influences outcomes, the choice warrants consideration. Possibilities for feature selection are illustrated in Figure 4.

A simple approach is removing one feature per sample. This enables assessing single feature relevance in model decisions, but downplays the complex relationships seen when multiple features are removed simultaneously.

¹<https://scikit-image.org/docs/stable/api/skimimage.segmentation.html>

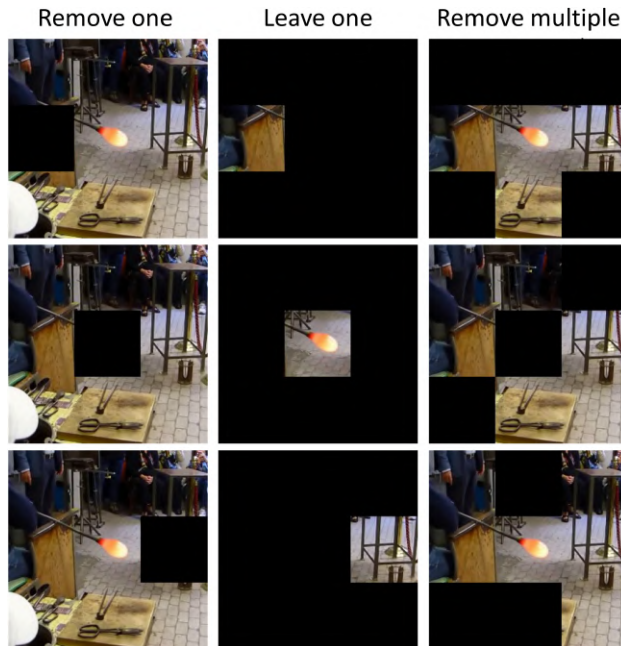


Figure 4: Feature selection steps. Individual frames are displayed, although the regions span subsequent frames. Left to right, samples with a single feature removed, with all but one removed, and with about half the regions removed. For visualization, 3D grid-based segmentation and black fill were employed to remove regions.

Methods employing this technique include occlusion sensitivity [21] and one of its adaptations to video, AOSA-SGL [12], as well as LOCO [27]. An alternative is to remove all but one feature, as with univariate predictors [28].

Alternatively, removing multiple features simultaneously can identify significant feature sets. However, this poses challenges in uncovering feature relevance. LIME [8], RISE [9], and Kernel SHAP [10] adopt this strategy. In LIME and RISE, each region is removed or retained with probability $p = .5$, thereby occluding about half the regions. This minimizes the sample requirements for an accurate approximation of feature relevance. It is of course possible to alter p to regulate the number of occluded regions or adopt a hybrid method, although this may increase the number of perturbed samples. Kernel SHAP, for instance, follows this strategy by employing a mix of samples with different numbers of occluded features than with uniform occlusions.

3.3. Sample Selection

To determine the significance of features, removal-based methods use perturbed input samples to correlate model prediction differences with each feature. Each explanation requires a specific number of model passes, corresponding to the number of samples, which may vary among explanation methods. Often treated as a hyperparameter, the number of samples impacts the quality of explanation. Achieving a balance between estimation accuracy and computational efficiency is vital: The explanation should be insightful yet not burdensome on the computational load.

Certain methods, such as occlusion sensitivity [21], predetermine the number of samples based on pixel count and occlusion kernel size. This is also the case with LOCO [27] and univariate predictors [28] which rely on feature count. Conversely, LIME [8], RISE [9], or Kernel SHAP [10] require a user-defined sample count. Generally, increasing the sample count improves the relevance quality. This is evident using Kernel SHAP, where testing all feature combinations is unfeasible; however, an approximation is achieved through a substantial sample count, leading to more accurate shapley value estimates. The appropriate number of samples depends on the feature count, hence video explanations generally require more samples than image explanations, unless the number of regions is reduced by enlarging their size. In the RISE experiment [9], 4,000 or 8,000 samples were used to compute explanations, contingent on the model; the public LIME image implementation defaults to 1,000 samples; and in the Kernel SHAP implementation, the default is 2,048 plus the number of features.

3.4. Feature Removal

For visual data, removing a pixel does not directly apply. Instead, we substitute (or occlude) the pixel with a color, effectively nullifying the information it carries. This section examines approaches for performing this occlusion, as demonstrated in Figure 5.

A simple method for occlusion is to fill regions with a color, for example, black (RISE [9]) or gray (occlusion sensitivity [21]). Since the result might include out-of-domain data, that is, colors not in the training dataset [33], an alternate method is to use the mean color of the occluded region, as with LIME [8]. Another technique is blurring, as described by Fong and Vedaldi [34]. More sophisticated methods, such as image inpainting [35], replace regions in a more intelligent manner such as eliminating information for class recognition while preserving colors in the image or video. Moreover,

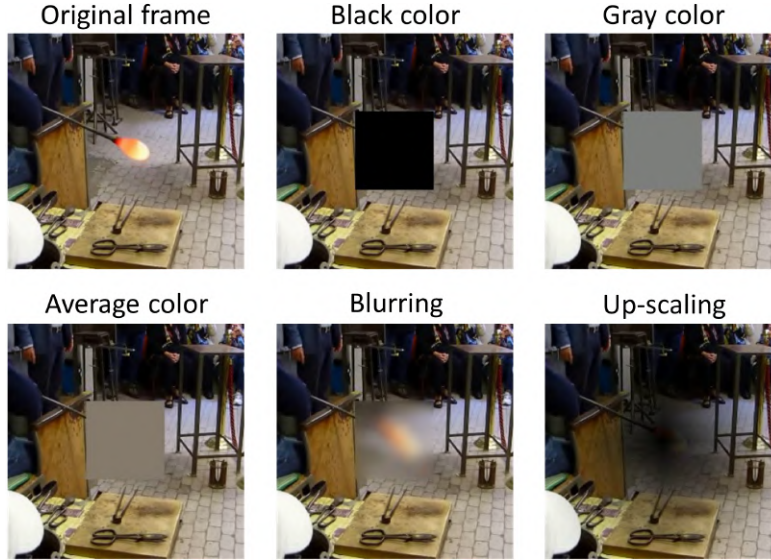


Figure 5: Feature removal methods. As labeled, original input, occlusion with black, gray, average color, uniform blur filter, and up-scaled black. Only a single frame is displayed, however, segmentation uses a 3D grid (i.e., across frames). For clarity only one region is occluded in each sample.

it is possible to use transparency (i.e., alpha) within the mask rather than entirely removing a region. An example is RISE [9], where a low-resolution grid is constructed and scaled up to produce the transparency gradient. This technique creates smoother occlusion and avoids sharp edges.

3.5. Visualization

The final step in the explanation process is visualization. Although, it might appear straightforward, visualization significantly impacts user comprehension of the explanations [36]. Visualization methods vary based on the explanation technique employed, as explored in this section.

A divergence among explanation techniques is whether relevance is solely positive or can also include negative values. Positive relevance occurs when removing a region diminishes confidence in a class, indicating the class’s necessity for accurate class prediction. Conversely, negative relevance occurs when removing a region increases confidence in a class, which means that the class mislead the model when present. Methods like LIME [8] and SHAP [10] include both positive and negative relevance, while RISE [9] and occlusion sensitivity [21] only produce positive relevance. Visualizing positive

relevance often uses a heat map with color representing the region’s importance. When incorporating both positive and negative relevance, distinct colors may differentiate them. For instance, LIME uses green and red for positive and negative relevance, respectively. An alternative approach focuses individually on positive or negative relevance and uses heat maps for visualization.

To enhance visualization, histogram stretching can be applied to re-scale relevance within a color space, aiding in the differentiation of relevance across regions. Additionally, rather than displaying all regions, a subset can be chosen, such as the n most relevant regions, regions surpassing a minimum relevance threshold, or regions starting from the most important and incrementally accumulating relevance until a set threshold is reached. These methods simplify and refine the explanations by mitigating noise.

Explanations can merge with the input image or video, affording a direct view of the relevance of each pixel. This is possible through alpha compositing; that is, blending explanations with the input data using transparency as a weighting. Alternatively, when magnitude is less significant, specific pixels in the n most relevant regions can be altered. However, care is warranted in considering the influence of input data on interpretation, as this can lead to color variations that incorrectly imply relevance. For instance, if relevance is displayed in green and the scene includes green elements, the merge might be ambiguous. This is addressed by adjusting colors for relevance display or by applying higher transparency to input data and lower transparency to explanations, accentuating explanation colors over the background.

Table 2 summarizes options to consider at each stage of the explanation process. Note that for certain steps, such as visualization, the options are not mutually exclusive.

Table 2: Options for Each Explanation Stage

1. Segmentation	2. Feature selection
• 3D grid • 3D superpixels • 2D segmentation with optical flow	• Removal of n regions • Remove with probability p • Fixed number of features removed • Removal of features based on a distribution
3. Sample selection	4. Feature removal
• Fixed value • Dependent on the number of features	• Fixed color • Mean color • Mean color of the removed region • Blurring • Upscaling • Inpainting
5. Model behavior	6. Summary technique
• Prediction • Prediction loss • Dataset loss	• Mean confidence of feature when included • Difference unperturbed and perturbed features • Shapley value • Linear model
7. Visualization	
• Heatmap to visualize magnitudes • Preserve/remove negative relevance • Display only the n most important regions • Minimum relevance threshold • Maximum accumulated relevance threshold • Histogram stretching • Alpha composing	

4. Methodology

Following our REVEX framework (see Figure 1), we extend removal-based explanation methods to video data. Three models form the basis for our explanations. These were trained on three dissimilar datasets to yield a range in predictions. Thus our evaluation spans different models, environmental contexts, and classes.

4.1. Networks

The networks used in our experiment are TimeSformer [37], TANet [38], and TPN [39]. These were chosen due to their robust performance in action classification tasks and their user-friendly implementation within MMAction2 [40], an open-source PyTorch-based toolbox for video comprehension. Their specifications are summarized in Table 3. The top-1 and top-5 accuracy metrics are from MMAction2 [40] and were computed on the Kinetics 400 dataset with pre-training on ImageNet.

Table 3: Network Specifications and Performance Benchmarks

Network	Backbone	Resolution	Frames	Crops	Accuracy	
					top-1	top-5
TimeSformer	TimeSformer (divST)	224×224	8	3	77.69	93.45
TPN	ResNet50	224×224	80	3	76.74	92.57
TANet	ResNet50	224×224	80	3	76.25	92.41

TimeSformer adapts the vision transformer architecture [41] to incorporate spatio-temporal feature learning via frame-level patches. We used the “divided attention” version, which demonstrates optimal performance by independently computing spatial and temporal attention.

TANet encompasses a temporal adaptive module (TAM) embedded in a 2D CNN architecture. It bears minimal overhead while capturing short- and long-term information through a two-level adaptive scheme. To achieve this, the TAM uses a local temporal window and a global aggregation weight.

TPN (temporal pyramid network) integrates into 2D or 3D backbones and enhances performance for classes with temporal variation. TPNs extract spatial, temporal, and informational features which are subsequently rescaled and concatenated.

4.2. Datasets

The Kinetics 400 [42], UCF101 [43], and ETRI-Activity3D [44] datasets were used to assess the explanations on video data. The goal was to obtain divergent results in both the evaluations and explanations, as the datasets have different characteristics.

Kinetics 400 contains 306,245 videos from YouTube containing 400 human action classes with a minimum of 400 clips per class. The videos include human-human and human-object scenarios with a range of individuals, environments, and objects. Moreover, the videos include camera movements and scene cuts, thereby adding complexity to the data.

UCF101 contains 13,320 videos from 101 classes, all sourced from YouTube. The videos are diverse in camera motion, object appearance and pose, object scale, and illumination. There are five class types: human-object interaction, body-only motion, human-human interaction, playing musical instruments, and sports. Additionally, UCF101-24, a subset of this dataset with just 24 classes, includes action detection annotations from the THUMOS-2013 challenge [45]; it was used to evaluate explanations.

In contrast, ETRI-Activity3D is a smaller set of 55 activity classes from 112,620 videos. The videos capture daily activities of 100 individuals, with half the participants above 64 years old and recorded in their home from various rooms and eight perspectives. Apart from the RGB videos, depth frames and skeleton joints data are also available. The recordings used fixed cameras and do not include scene cuts; they offer a controlled and stable setting for evaluation.

4.3. Training

For training the TimeSformer, TANet, and TPN networks we used a computer with an NVIDIA 4090 GPU and an i9 9900KF CPU, kindly provided by the local university. Training and evaluation used MMAAction2 [40], an open-source toolbox for video understanding based on PyTorch and part of the OpenMMLab project.

The models were initialized with pre-trained weights from the Kinetics 400 dataset, and further trained for 10 epochs on the ETRI-Activity3D and UCF101 datasets using K-cross validation with $K = 5$. Training the models on the Kinetics 400 dataset was unnecessary, as pre-trained weights are publicly accessible online.

Table 4: Step Choices by Video Explanation Method

Step	LIME	K.-SHAP	RISE	LOCO	UP	SOS
Segment (Regions)	SLIC (≈ 200)	SLIC (≈ 200)	3D grid ($4 \cdot 7^2 = 196$)	SLIC (≈ 200)	SLIC (≈ 200)	3D grid (1183)
Feat. Sel.	Multiple	Multiple	Multiple	All but one	Only one	All but one
Sample Sel.	1000	1000	1000	≈ 200	≈ 200	(1183)
Feat. Rem.	Blur	Blur	Blur	Blur	Blur	Blur
Model beh.	Prediction	Prediction	Prediction	Prediction	Prediction	Prediction
Summary	Linear model	Shapley value	Mean ¹	Difference ²	Difference ²	Pred. value ³
Visualiz.	Heat map	Heat map	Heat map	Heat map	Heat map	Heat map

¹ Mean when included² Difference with base case: no perturbations for Video LOCO and all perturbed for Video UP³ Prediction when occluded

4.4. Explanation Methods

Our experiment used six explanation methods: LIME [8], Kernel SHAP [10], RISE [9], LOCO [27], univariate predictors [28], and occlusion sensitivity [21]. While LIME, Kernel SHAP, RISE, and occlusion sensitivity work with image inputs and require minimal change for video input, LOCO and univariate predictors needed substantial modifications for video inputs and provide local rather than global explanations. Table 4 details the adaptations for each explanation method to work with videos and itemizes the choices for the steps in feature-removal. Steps and method names are abbreviated for space reasons.

To adapt LIME and Kernel SHAP to videos, the only modification was segmentation, where pixels across frames are used to compute superpixels. For RISE, we added a dimension to the 2D grid for temporal information. The LOCO (leave-one-component-out) explanation method originally requires retraining the model multiple times, excluding a specific feature each time to determine global feature relevance. However, in visual data, global explanations often lack precision due to variability in camera perspective and object positions. To address this, we independently computed the difference between occluding or including each feature (or region), thus yielding the local relevance of each feature. This approach also avoids the need for repeated retraining, a process that is particularly time-intensive with large datasets. The univariate predictors method was adapted to explain individual predictions (local) rather than the full dataset (global). The relevance of a variable (or region) was found by occluding all but that variable and observing the

prediction. For occlusion sensitivity, we adapted the 2D occlusion kernel to three dimensions and reduced the passes by using a subset of occlusions. The occlusion process was regulated by a stride value: the number of occlusion possibilities in each dimension. This allowed us to manage computational complexity effectively.

For methods using superpixels for segmentation (i.e., LIME, Kernel SHAP, LOCO, and univariate predictors), we used SLIC [23] due to its ease of extrapolation to three dimensions. We experimentally set the number of segments to approximately 200 and used $5\times$ this value as the number of samples for methods requiring this parameter (LIME, Kernel SHAP, and RISE).

We experimentally selected a Gaussian blur to apply perturbations to the input, similarly to Fong and Vedaldi [34]. Occlusion with a specific color (e.g., black) introduced too much noise to the video, preventing the identification of a class independently of the occluded region. Instead, we first computed a blurred version of the input and introduced its pixels to the perturbed regions. This approach is better in avoiding out-of-domain data [33], since perturbations are generated using existing data. Additionally, the masks applied to the input perturbations were also blurred, with a smaller kernel size, to get a “fade-in” effect; this allowed hard edges to be removed from the perturbed samples. A fade-in effect was also used in visualizing the explanations, as faithful to the shape of each region when occluded.

Figure 6 presents perturbations and explanations for each feature removal method. The method names are prefaced with “Video” to reflect their new context and simplified for brevity: Video LIME, Video Kernel-SHAP, Video RISE, Video LOCO, Video UP (univariate predictors), and Video SOS (sampled occlusion sensitivity, to highlight sampling with a stride).

To compare the methods, we selected two methods that are model-specific (Grad-CAM [4] and saliency tubes [6]) and two that are model-agnostic (AOSA and AOSA-SGL [12]). Gradient-weighted class activation mapping (Grad-CAM) extends to video and uses gradients from the prediction of a class to weight features from the last layer of a convolutional neural network (CNN). Saliency tubes, although inspired by Grad-CAM, was specifically designed for 3D CNNs to work with video. Finally, Adaptive occlusion sensitivity analysis (AOSA) was proposed as an extension to video work by Zeiler and Fergus [21]. It uses optical flow to extend 2D patches through the temporal dimension.

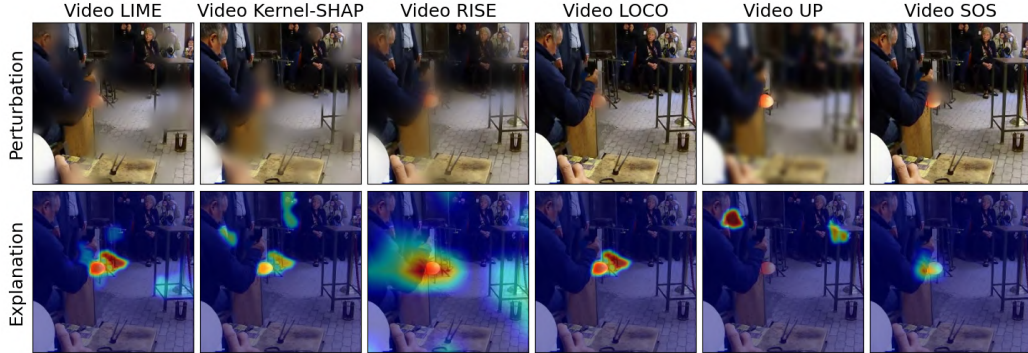


Figure 6: Example explanations by feature removal method. Top row: perturbed sample. Bottom row: saliency map. Explanations were computed for the TANet model on the Kinetics 400 dataset for “blowing glass.” While only a single frame is shown, perturbations and explanations extend to the entire video sequence.

4.5. Explanation Process

To evaluate the explanation methods, we randomly selected 30 videos from each dataset discussed in subsection 4.2. To ensure consistency, we only included videos that were correctly classified by the three networks in the experiment.

For each dataset, the 30 videos were passed through the three networks described in subsection 4.1, with prediction explanations generated using each method detailed in subsection 4.4 (the six proposed methods, plus four existing ones). This resulted in $30 \times 3 \times 3 \times 10 = 2,700$ explanations.

For consistency, negative relevance was removed from the explanations, as some lack both positive and negative relevance values. For visualization, heat maps were used for all methods, with important regions represented by warmer colors. Additionally, histogram stretching was applied to maximize the color range for clearer interpretation.

4.6. Evaluation of the Explanations

To assess the quality of the explanations, we mined the literature to find strategies for evaluating explanation methods. In this section, we explain these strategies.

4.6.1. The Deletion and Preservation Games

Deletion and preservation games, introduced by Fong and Vedaldi [34], are used three ways: to determine minimal masks at target thresholds, to

compute the area under the curve (AUC) of varying thresholds, and to estimate the average drop in prediction. Figure 7 illustrates our application of these methods to the full set of 2,700 explanations.

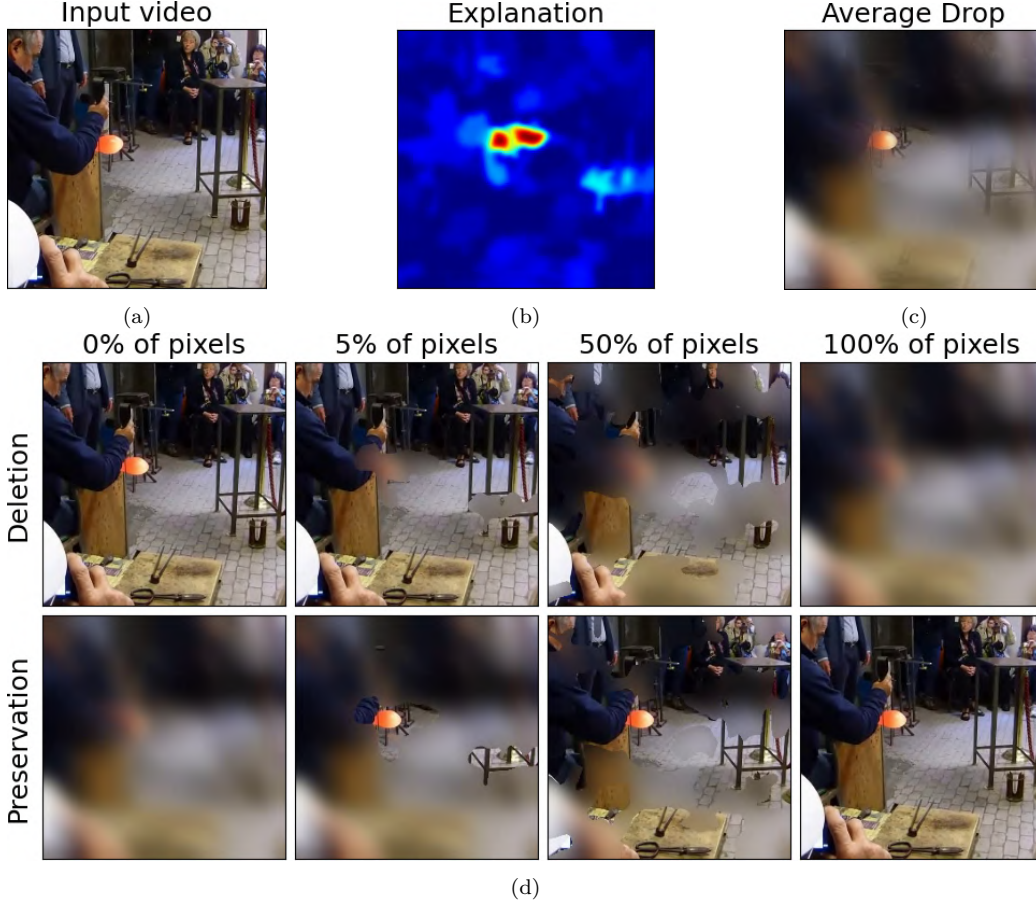


Figure 7: Perturbations introduced by deleting or preserving regions in the input video, computed on Video LIME explanations for TANet trained on Kinetics 400. Only one frame is shown for simplicity. (a) Input video on “blowing glass.” (b) Saliency map, with warmer colors showing greater relevance. (c) Perturbation using the average drop method, with less relevant regions more perturbed. (d) Deletion (top row) and preservation games (bottom row), where regions are removed or preserved according to their relevance. Columns represent perturbations on 0%, 5%, 50%, and 100% of relevant pixels.

For minimal masks evaluation, the goal is to identify the smallest set of regions that either obstruct the network in accurately recognizing the correct class (deletion) or assist the network for the correct classification (preserva-

tion) [34]. For this, minimal masks at target prediction thresholds are identified with the pixels in the masks counted and expressed as a percentage. Models will differ in resolution and frame count.

Inspired by minimal masks evaluation, Petsiuk et al. [9, 46] proposed using the area under the curve (AUC) of the deletion and preservation games as a metric. AUC is applied to the prediction confidence as pixels are deleted or preserved. Unlike the minimal masks evaluation, deletion and preservation AUCs do not use a target prediction threshold but rather summarize the results across thresholds. AUC values range between 0 and 1 and should be minimal for deletion (with a minimum value of 0) and maximal for preservation (with a maximum value of 1). Petsiuk et al. referred to the two metrics as “deletion” and “insertion” metrics, so this naming is adopted for the remaining of the document.

Chattopadhyay et al. [47] proposed a variation on deletion and preservation called “average drop” that measures the average decrease in the prediction when using perturbed and unperturbed video data. Average drop uses the saliency map to determine the regions to perturb. Regions with more relevance are modestly perturbed, while regions with less relevance are highly perturbed. Only one pass through the network is required. Average drop is minimal for the explanation method to get a good score.

We utilized the “deletion” and “insertion” metrics proposed by Petsiuk et al., as they improve upon the original minimal mask evaluation by Fong and Vedaldi [34] by eliminating the need to select specific thresholds. In addition to these metrics, we included the average drop metric in our experiment. In the three metrics, we utilized a Gaussian blur to remove regions in the deletion metric and as the default value for regions not yet included in the insertion metric. This choice ensures consistency with the perturbations for the explanations, which were generated through the same blurring process.

4.6.2. Weakly Supervised Object Localization

By extracting regions the model deems crucial, it is possible to build a mask for object detection and segmentation without training on a dataset with such labels. This approach is often used [2, 48, 49, 34, 4, 47] but it requires a mask or bounding box for ground truth. Consequently, it was computed only on the 900 explanations from the UCF101-24 dataset.

In line with prior studies, we used the intersection over union (IoU) metric and a threshold of 0.5 to evaluate the object localization error for each method [34, 49, 48, 2, 4]. To compute it, we binarized the saliency maps through

thresholding, and selected the threshold that optimized the IoU by sampling across $[0, 1]$ in 0.05 increments. Additionally, following Chattopadhyay et al.’s methodology [47], we used the mean IoU for each method, rather than relying solely on the 0.5 threshold. For the remaining of this paper, these two metrics are referred to as “IoU accuracy” and “average IoU”.

Another approach for evaluating explanation localization is the pointing game [49, 4, 9, 46], which determines if the maximum point on the saliency map falls within the ground truth bounding box. If it does, it is considered a hit; otherwise, it is a miss. The localization accuracy is then calculated as $Acc = \frac{\#Hits}{\#Hits + \#Misses}$, as defined by Zhang et al. [49].

The two approaches are displayed in Figure 8.

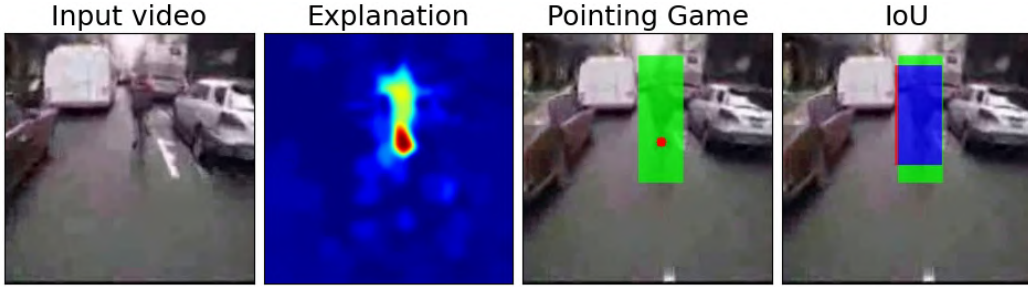


Figure 8: Comparison of pointing game and intersection of union (IoU). Left to right: input video, saliency map, pointing game evaluation, and IoU evaluation. Only the frame for maximum relevance is displayed. In the saliency map, relevant regions are in warmer colors. For the pointing game evaluation, the ground truth bounding box is green, while the point of maximum relevance is red. For the IoU evaluation, true positives are blue, false negatives are green, and false positives are red.

5. Results

In this section, we present the results of the experiment using the explanation methods outlined in Section 4.6. To provide a deeper understanding beyond numerical outcomes, we include additional examples in Appendix A across explanation methods, networks, datasets, and classes.

5.1. Explanation Time

This section presents the average time per video for each explanation method. Figure 9a highlights variations in explanation time by network architecture. The significance of time varies by the context in which XAI

techniques are employed. Since the techniques are typically applied in developing DL models rather than in real-time use, time is generally not a critical concern.

TANet and TPN, which use larger input sizes than TimeSformer, require more inference time. This impacts the methods that necessitate multiple forward passes, such as the proposed approaches. Conversely, methods like Grad-CAM and saliency tubes exhibit lower computation times for TANet and TPN, as they require only a single forward and backward pass through the network. However, these methods cannot work with TimeSformer due to architectural incompatibility. All methods were evaluated on down-scaled versions of the videos ($16 \times 112 \times 112$) to ensure a fair comparison. This was necessary due to the higher memory requirements of AOSA and AOSA-SGL², which preclude using full-size input videos.

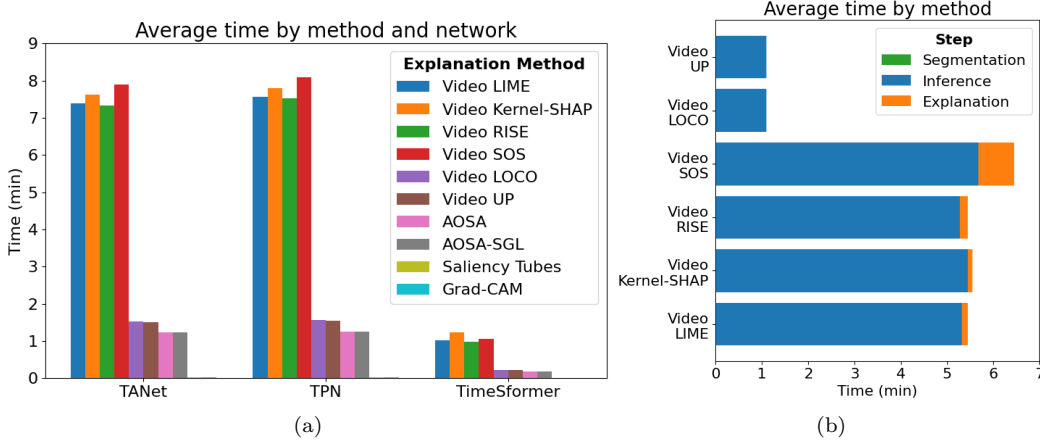


Figure 9: Time by method and network. Input was a single video of resolution $16 \times 112 \times 112$ (frames $\times$ width $\times$ height). (a) For comparison, existing methods are included. Bars for saliency tubes and Grad-CAM are barely visible for TANet and TPN as time was under five seconds, and are absent for TimeSformer. (b) Time for the proposed methods by segmentation time (primarily for SLIC computation), inference time (depending on network and sample size), and explanation time (depending on video and sample size).

Figure 9b provides a breakdown of the time per video, considering all datasets and networks for the proposed explanation methods. Time is split by step; that is, segmentation, inference, and explanation. Although near zero for the given resolution, segmentation time is appreciable for methods

²<https://github.com/uchiyama33/AOSA>

that utilize SLIC to partition the video into regions (all proposed methods except Video SOS and Video RISE, which rely on grid-based segmentation). Inference time varies with the network and scales linearly with the number of samples, which differs among explanation methods. Video-UP and Video-LOCO, which require only about 200 samples, needed far fewer passes compared to the remaining methods which required roughly 1,000 passes. Finally, “explanation time,” encompasses various steps such as feature removal, relevance summary, and visualization. Among these, feature removal is the most computationally expensive, with complexity influenced by the operations involved, the video size, the number of samples, and the number of regions occluded per sample. Video UP and Video LOCO demonstrated faster feature removal, as they occlude only one region at a time. In contrast, Video Kernel-SHAP, Video LIME, and Video RISE are more time-consuming as they occlude multiple regions simultaneously. Notably, Video-SOS demonstrated the poorest performance.

5.2. Deletion and Preservation Games

Results by explanation method using the AUC of the deletion and preservation games and incremental thresholds are presented in Table 5. The best performance for each network appears in bold. Averaging was not possible for saliency tubes and Grad-CAM, as they are incompatible with TimeSformer’s architecture. For deletion, lower values are better as they imply a greater drop in performance when relevant regions are removed compared to irrelevant ones. Conversely, for insertion, higher values are desired as they reflect a greater improvement in performance when relevant regions are added back.

The results for deletion and insertion show similar trends. The Video LIME method performed best overall, followed by Video RISE, Video Kernel-SHAP, and saliency tubes, though rankings varied depending on network and dataset. The poorest results were observed with Video UP, AOSA, AOSA-SGL, and Video SOS. More insight into these two metrics is found in Figure B.16 in the appendix. The best performance for deletion and insertion was obtained for explanations on the ETRI-Activity3D dataset, with minimum deletion values around 0.1 and insertion values near 0.95. Interestingly, Video LOCO demonstrated relatively strong performance for insertion, specifically for the ETRI-Activity3D dataset.

Similar to deletion, lower average drop values are favorable as they indicate less decline in performance when removing regions based on relevance (i.e., important regions are preserved more often than less important ones).

Table 5: Deletion and Insertion Scores by Explanation Method and Network

Explanation Method	Deletion ($\downarrow$)			Insertion ($\uparrow$)		
	TANet	TPN	TimeSf.	TANet	TPN	TimeSf.
Video LIME	0.1694	0.1496	0.1949	0.7654	0.7646	0.8120
Video Kernel-SHAP	0.2166	0.1842	0.2741	0.7179	0.6912	0.7400
Video RISE	0.2171	0.1788	0.2278	0.7347	0.6703	0.7542
Video SOS	0.3800	0.2580	0.3793	0.5923	0.5712	0.6390
Video LOCO	0.3123	0.2421	0.3284	0.6353	0.6568	0.7175
Video UP	0.4294	0.4042	0.5017	0.5324	0.4607	0.5917
AOSA	0.3675	0.3503	0.4888	0.6192	0.5756	0.6582
AOSA-SGL	0.3786	0.3632	0.4992	0.5911	0.5709	0.6281
Saliency tubes	0.2190	0.1698	-	0.7167	0.7077	-
Grad-CAM	0.3177	0.1971	-	0.6416	0.6885	-

Table 6 presents these results by explanation method and network, with the best performance in bold. Note that saliency tubes and Grad-CAM are not applicable to the TimeSformer network. Figure B.17 in the appendix displays broader results across explanation methods and datasets. Although Video LIME fared best for deletion and insertion (Table 5), the top performer for average drop was Video RISE with values below 7% across all cases, and even under 1% for the TimeSformer network.

Table 6: Average Drop Scores by Explanation Method and Network

Explanation Method	Average Drop (% , $\downarrow$)		
	TANet	TPN	TimeSf.
Video LIME	20.51	31.88	11.19
Video Kernel-SHAP	27.22	37.73	18.60
Video RISE	3.11	6.10	0.83
Video SOS	37.93	40.16	26.76
Video LOCO	73.14	76.20	61.38
Video UP	69.19	80.98	66.70
AOSA	34.15	22.13	21.99
AOSA-SGL	41.22	25.02	25.20
Saliency tubes	23.50	19.10	-
Grad-CAM	29.71	24.05	-

Video SOS performed well on the Kinetics 400 dataset but demonstrated

a significant drop in performance for the other two datasets. Video LIME achieved particularly strong results on the TimeSformer model and maintained consistent and fair results across the other models. AOSA, AOSA-SGL, saliency tubes, and Grad-CAM all performed well on the TPN model. In contrast, Video LOCO and Video UP consistently delivered the poorest results, showing a significant gap in performance compared to the other methods.

5.3. Weakly Supervised Object Localization

Accuracy results for weakly supervised object localization (i.e., pointing game, IoU accuracy, and average IoU) were computed solely on the UCF101 dataset, as it is the only dataset with bounding box annotations. The results are given in Table 7 and Table 8 with the best results in bold. Again, saliency tubes and Grad-CAM are not applicable to the TimeSformer network. Overall, the pointing game provided the highest performance with accuracy between 20% and 60%. In contrast, average IoU accuracy was between 10% and 20% and IoU accuracy was below 10% for almost all cases.

Table 7: Pointing Game Accuracy by Explanation Method and Network

Explanation Method	Pointing Game Accuracy (%)		
	TANet	TPN	TimeSf.
Video LIME	33.33	33.33	23.33
Video Kernel-SHAP	36.67	30.00	16.67
Video RISE	40.00	33.33	33.33
Video SOS	40.00	53.33	26.67
Video LOCO	26.67	36.67	43.33
Video UP	33.33	23.33	20.00
AOSA	26.67	23.33	30.00
AOSA-SGL	23.33	30.00	33.33
Saliency tubes	30.00	26.67	-
Grad-CAM	13.33	10.00	-

Pointing game performance varied considerably across models. For the TANet network, Video RISE and Video SOS achieved the highest accuracies (40.00%), followed by Video Kernel-SHAP, Video LIME, and Video UP. On the TPN network, Video SOS stood out with 53.33% accuracy, significantly

Table 8: IoU Accuracy and Average IoU by Explanation Method and Network

Explanation Method	IoU Accuracy (%) ¹			Average IoU (%)		
	TANet	TPN	TimeSf.	TANet	TPN	TimeSf.
Video LIME	10.00	10.00	6.67	17.10	17.68	15.67
Video Kernel-SHAP	6.67	6.67	6.67	16.22	16.48	15.72
Video RISE	6.67	6.67	6.67	15.30	15.87	16.23
Video SOS	0.00	3.33	3.33	11.78	12.03	13.00
Video LOCO	10.00	10.00	16.67	16.54	16.23	18.15
Video UP	6.67	6.67	3.33	15.28	16.86	15.33
AOSA	6.67	6.67	3.33	16.63	14.08	15.38
AOSA-SGL	3.33	6.67	3.33	13.51	13.37	14.52
Saliency tubes	6.67	6.67	-	17.00	16.78	-
Grad-CAM	0.00	6.67	-	14.66	15.30	-

outperforming the next best method, Video LOCO at 36.67%. In the TimeSformer network, the highest performance was observed with Video LOCO at 43.33% accuracy, followed by Video RISE and AOSA-SGL both at 33.33%. Interestingly, some methods that underperformed in the deletion and insertion tasks, like Video SOS and Video LOCO, demonstrated strong or even the best performance in the localization task.

When evaluating the methods using IoU-based metrics and including both IoU accuracy and average IoU, the results were consistent across methods. Notable exceptions were Video LOCO, which exhibited superior performance, particularly on the TimeSformer model, achieving an IoU accuracy 10% higher than the other methods. Conversely, Video SOS consistently showed the poorest results across both metrics for all test cases.

6. Discussion

In this section, we discuss removal-based methods, assess the strengths and weaknesses of the six proposed methods by analyzing the experimental results, and highlight limitations of evaluation metrics in the literature.

6.1. Removal-Based Methods

For removal-based explanations for visual data, it is essential to consider the steps during the explanation process. This study explored individual steps and discussed the parameters that bear on segmentation, occlusion, and visualization technique. This analysis aids in classifying existing explanation

methods and paves the way for improvements by combining strengths from different methods and steps.

In our view, the components of explanation methods such as segmentation, feature removal, and summary technique merit separate analysis and evaluation to improve each method. For instance, the LIME approach of training a linear model to find the relevance of each feature could apply to up-scaled grid perturbations generated by RISE. Because there are many explanation methods, this approach offers a better understanding of methods, with follow-up work leveraging the best of it and disregarding the rest. Following the same idea, segmentation merits deeper study, for example, on the impact of the number of regions, clustering for super-pixels, or the relevance of temporal information. Furthermore, segmenting in a single dimension can potentially lead to explaining the dimension separately [11]. To fairly compare the proposed methods, our experiment enforced equal conditions for the methods, such as the number of features, samples, occlusion type, and visualization. However, the selection of hyper-parameters was performed heuristically, leaving the investigation of optimal values for future work.

A notable outcome in the experiment was the improvement in explanations using blurring to introduce perturbations in the input space (cf. using a specific color, such as black). This is akin to introducing out-of-domain data when using a specific color [33], while blurring has the advantage of using data already present. Furthermore, the way perturbations are added is important: Using “fade in” leads to more faithful explanations and removes hard edges that could lead to artificial new shapes.

6.2. Proposed Methods: Strengths and Weaknesses

Six distinct video explanation methods were proposed, all adapted from existing approaches. Adapting LIME, Kernel-SHAP, and RISE to video was relatively straightforward, as these methods were initially designed for images, and therefore only required adding a temporal dimension during segmentation. The adaptation of Video SOS offers a natural and robust solution to handle camera movements and cuts, as compared to existing adaptations that rely on optical flow [12]. This enables the identification of relevant features in the temporal dimension. Furthermore, we introduced novel adaptations, namely Video LOCO and Video UP, to generate local explanations, extending their applicability to the visual domain beyond global explanations.

The primary limitation of removal-based approaches is the computational cost, as they require multiple perturbed samples and forward passes through the model. However, they are model-agnostic, meaning they apply to any model. This contrast is evident in Figure 9a, where Grad-CAM and saliency tubes could not work with the TimeSformer model, while removal-based methods could, though the latter required more computation. Conversely, the model-specific methods (Grad-CAM and saliency tubes) were almost instantaneous for the architectures they could be used on. While Video LOCO and Video UP are removal-based and faster, the cost was less robust results due to the smaller number of samples used.

In evaluating the explanation methods, we employed existing metrics categorized in two groups: the deletion and preservation games and weakly supervised object localization. In the deletion and preservation games, Video LIME and Video Kernel-SHAP performed best for deletion and insertion values, while Video RISE outperformed others in the average drop metric. Thus, these three methods effectively identified the key regions influencing model performance when removed or added. In contrast, Video SOS excelled in the weakly supervised object localization task, particularly in the pointing game, and Video LOCO achieved the best IoU accuracy. The other proposed methods demonstrated solid results in these metrics, particularly in the pointing game, often surpassing saliency tubes and Grad-CAM, indicating better localization of relevant regions, notably around human bounding boxes.

One significant finding was the low performance of the Video UP method, as reflected in multiple metrics. This is due to excessive perturbations in the explanations, where most samples had all but one region perturbed. As a result, these samples may fall outside the domain, failing to improve confidence in the correct class, even when crucial regions are not perturbed. Additionally, Video Kernel-SHAP explanations scored lower than Video LIME, likely due to the number of samples with over half the regions occluded, leading to potentially noisy or misleading explanations. Further investigation is warranted to enhance the performance of these two methods.

When comparing our proposed video sampled occlusion sensitivity (Video SOS) method to the two versions of adaptive occlusion sensitivity analysis (AOSA and AOSA-SGL) [12], we observed that Video SOS was better in the deletion metric (0.34 vs. 0.40 and 0.41, averaging results across datasets and models) and the pointing game (40.00% vs. 26.67% and 28.89%). However, it yielded poorer results in the average drop (34.95% vs. 26.09% and 30.48%)

and comparable results in the insertion metric (0.60 vs. 0.62 and 0.60), IoU accuracy (0.74% vs. 1.85% and 1.48%), and average IoU (12.27% vs. 15.36% and 13.80%). Note that for the deletion and average drop metrics, lower values are preferred. These results demonstrate the effectiveness of our approach, showing that simply adding the temporal dimension to the occlusion patch yields accurate explanations. This is in contrast to optical flow, which struggles with camera cuts and rapid movements, and completely disregards temporal relevance. To further improve Video SOS for explanation speed and performance, strategies to include perturbation of multiple regions concurrently are topics for future work, following the example of other methods, including AOSA [12].

Lastly, a strength of methods relying on SLIC-based segmentation, such as Video LIME, Video Kernel-SHAP, Video LOCO, and Video UP, is their ability to generate fine-grained explanations compared to methods like saliency tubes and Grad-CAM, which are constrained by the resolution of the final convolutional layer of the network. Additionally, they also surpass methods like Video RISE, Video SOS, AOSA, and AOSA-SGL, which rely on grid-based perturbations. In contrast, superpixel-based segmentation provides more intuitive perturbations, as it tends to include entire objects or uniformly colored regions in the same segment, leading to more interpretable explanations.

6.3. Main Problems Found With Existing Evaluation Methods of Explanation Methods

To enable a rigorous comparison of the explanation techniques, we adopted evaluation methods widely used in the literature. However, there are inherent limitations to consider during application. First and foremost, it is important to recognize that the deletion and preservation games and weakly supervised object localization metrics assess entirely different aspects of model explanations. As shown in previous sections, the methods that perform best on these two metrics often differ.

The deletion and preservation games evaluate how effectively the explanation methods identify the most relevant input regions. However, this evaluation relies on perturbations in the input space, which inadvertently affect model predictions regardless of the region perturbed. Therefore, we suggest computing models’ robustness to increasing levels of random perturbation and factor that into the deletion and insertion scores. Additionally, we believe the average drop metric may favor methods with more sparse saliency

maps, as these result in fewer perturbed regions, thereby leading to smaller decreases in model confidence. This could explain the superior performance of Video RISE on this metric compared to other methods, while it was not the case in other metrics.

In contrast, weakly supervised object localization evaluates the spatial alignment of relevant regions in the saliency map, which is primarily influenced by the network itself rather than the explanation method. For this reason, we argue that these metrics are more suitable for comparing trained networks than explanation methods. A clear example is Figure A.15, which depicts the “cricket bowling” activity. Here, the most relevant region is the wicket, which is outside the human bounding box. Although this explanation appears reasonable and is consistent across multiple methods, localization metrics would still yield low scores for these explanations. Moreover, we believe that IoU-based metrics overlook focused explanations while favoring more diffuse or widespread explanations. For instance, focusing solely on a person’s face may be sufficient for correct classification, but the face’s relatively small area compared to the bounding box (which encompasses the full body) yields a low IoU score. Conversely, the pointing game recognizes this as a valid explanation. Overall, we believe the pointing game is an appropriate metric for determining if models attend to the desired regions (e.g., it could be useful in identifying training flaws in healthcare applications). In contrast, IoU-based metrics are more appropriate for determining whether a trained model is suitable for weakly supervised object localization tasks.

7. Conclusions

As the applications of AI expand, the importance of explainable AI (XAI) techniques is increasingly evident, but while numerous approaches exist for image data, the video domain is insufficiently explored. Therefore, in this research, we decomposed the explanation process of removal-based methods into well-defined steps, specifically aimed at visual data. The steps are segmentation, feature selection, sample selection, feature removal, model behavior, summary technique, and visualization. This distinction in the pipeline is crucial when dealing with video data due to the added temporal dimension. Leveraging this framework, we adapted six removal-based explanation methods to the video domain, incorporating necessary adjustments in each case.

The evaluations used metrics for the deletion and insertion game and the weakly supervised localization task. The results were consistently good for the proposed methods across the different metrics. Specifically, Video LIME, Video Kernel-SHAP and Video RISE showed better performance for the deletion and insertion metrics than model-specific methods like Grad-CAM and saliency tubes, and other video-specific removal-based approaches, like AOSA. Video SOS and Video LOCO excelled in the localization metrics.

Decomposing the explanation process into well-defined steps presents opportunities for investigating the influence of each choice in the explanation pipeline, ultimately improving the explanation methods. Future research should combine aspects from different methods to derive optimal synergies and choices for each step to optimize the methods and better understand how different perturbations affect model predictions.

Appendix A. Examples of explanations

Figures A.10, A.11, A.12, A.12, A.14, and A.15 present examples of model explanations. The figures illustrate the explanations from all methods in this study for the TANet, TPN, and TimeSformer networks. Two classes from each dataset—ETRI-Activity3D, Kinetics 400, and UCF101—are displayed.

Each figure is organized as follows: Rows show the input video followed by rows for the six proposed methods and four additional methods for comparison. Columns are divided into three groups (one per network), each containing three columns (first, middle, last) for frames sampled at consistent intervals for comparison.

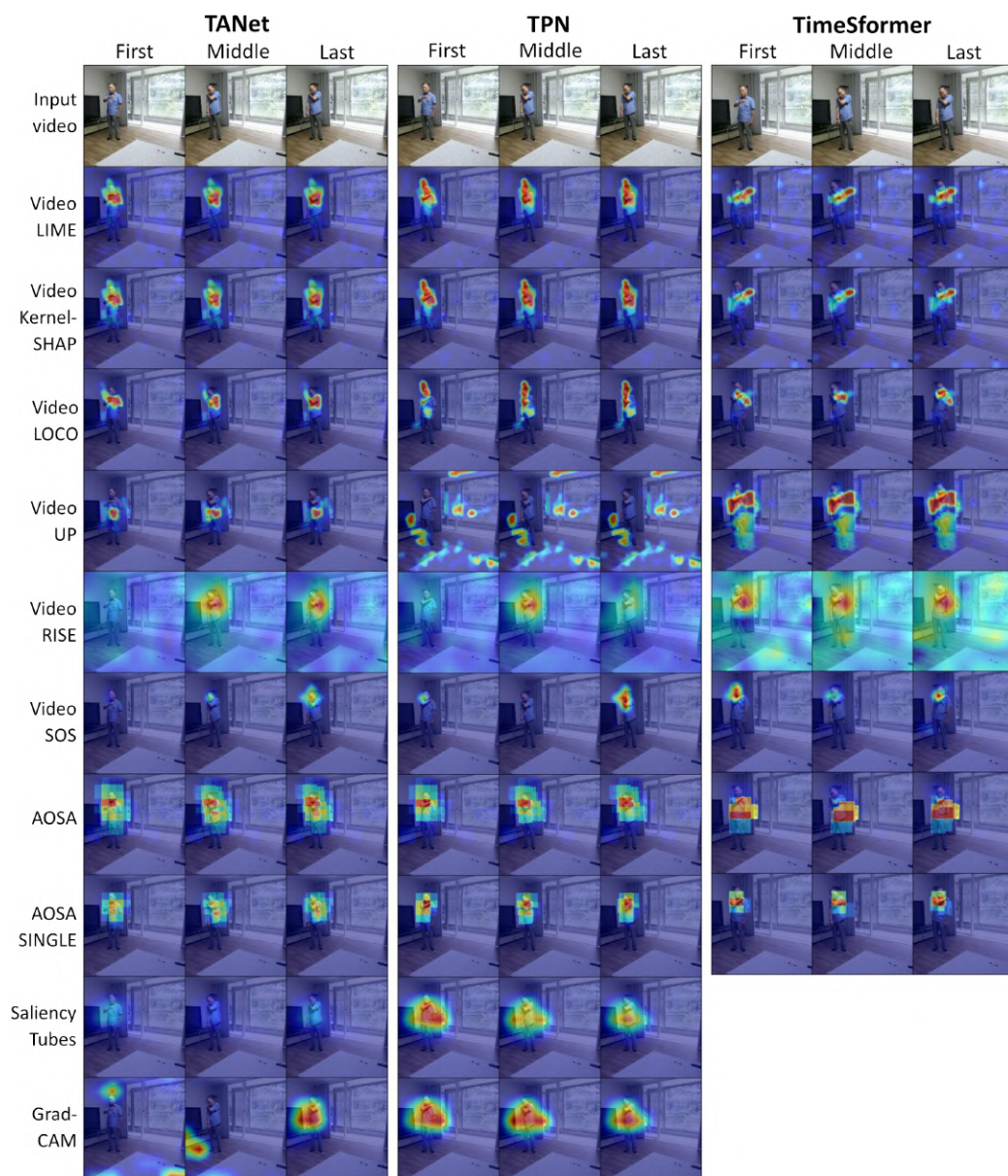


Figure A.10: Explanations for the ETRI-Activity3D class “massaging a shoulder oneself.”

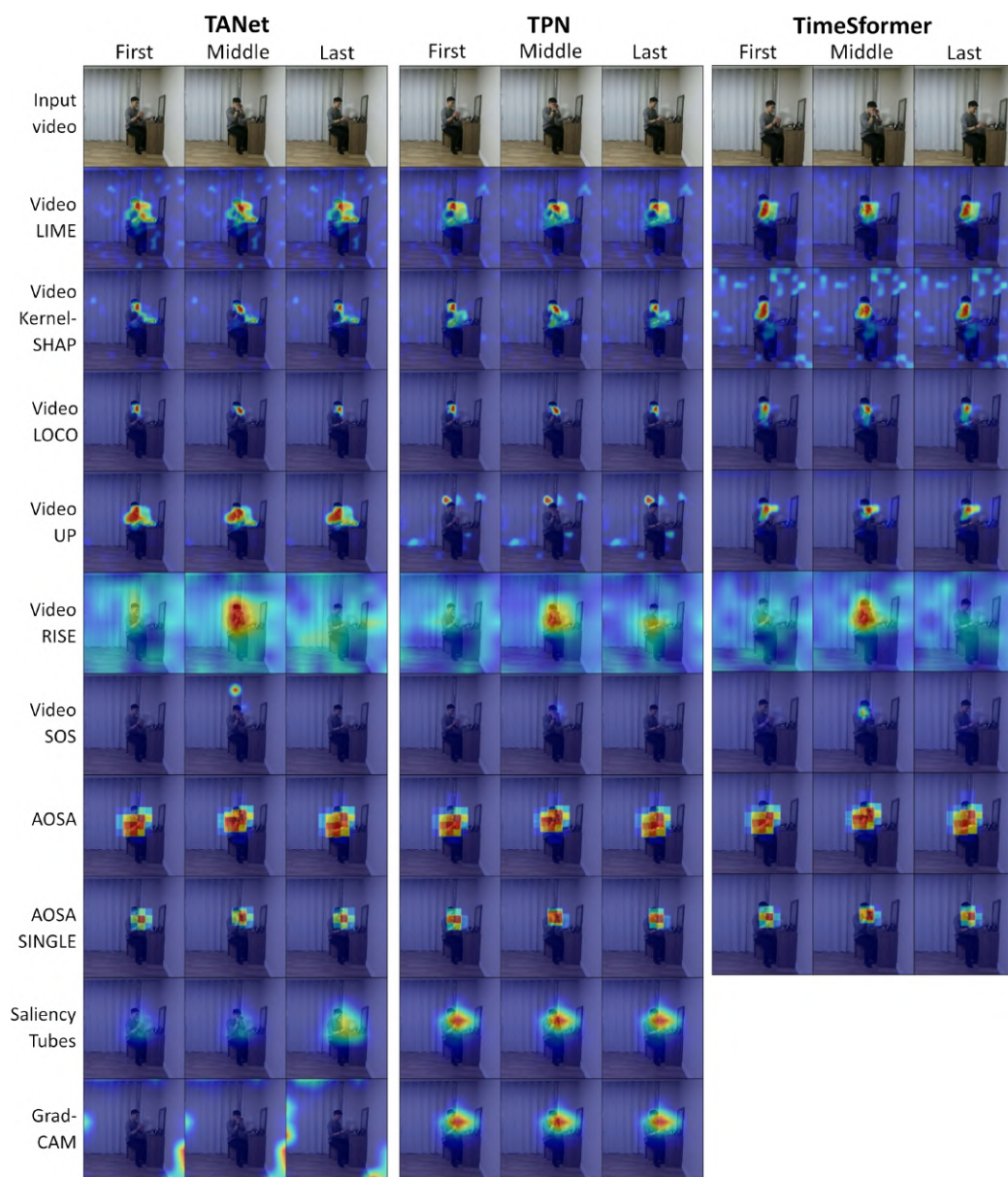


Figure A.11: Explanations for the ETRI-Activity3D class “putting on/taking off glasses.”

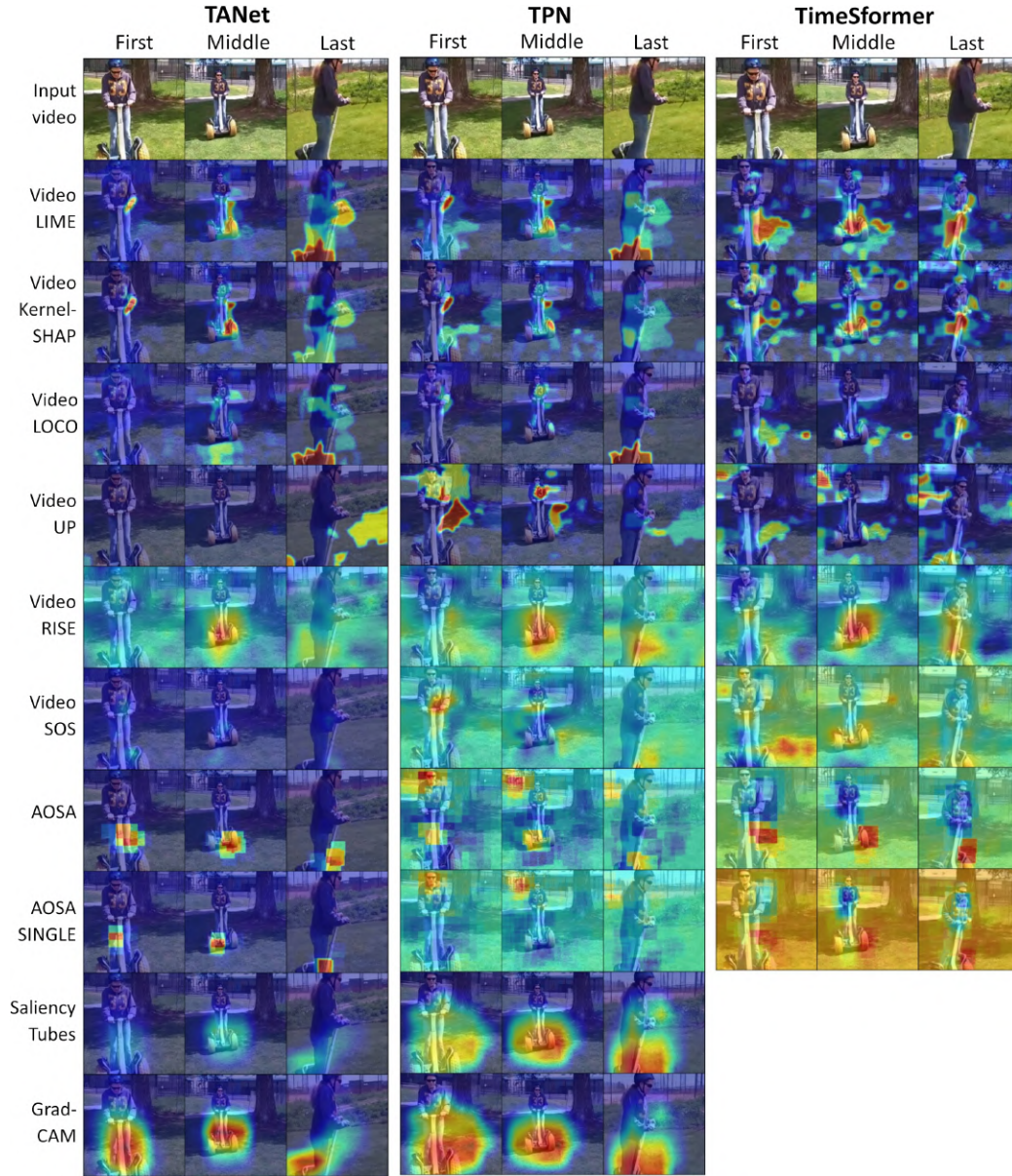


Figure A.12: Explanations for the Kinetics 400 class “using segway.”

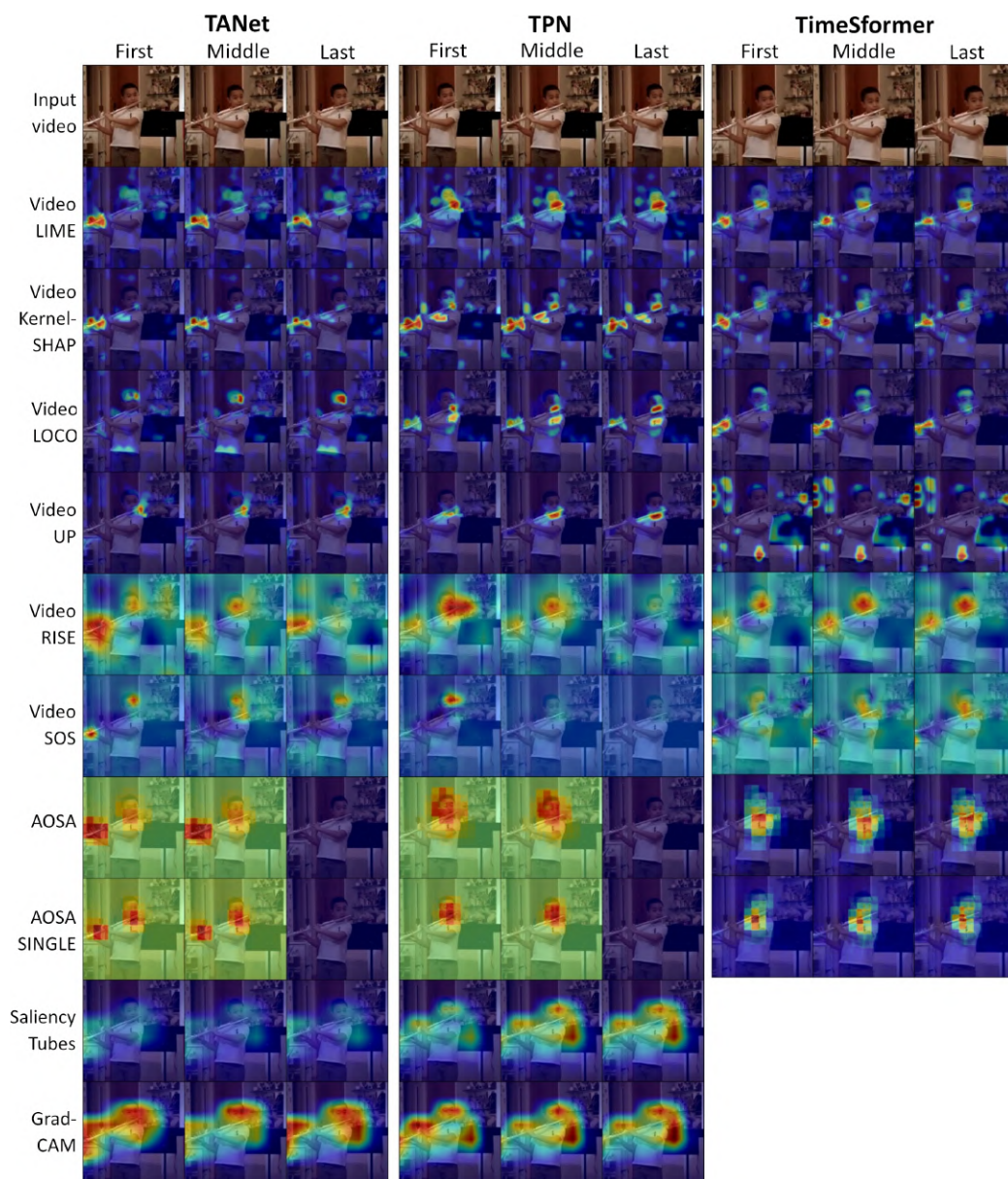


Figure A.13: Explanations for the Kinetics 400 class “playing flute.”

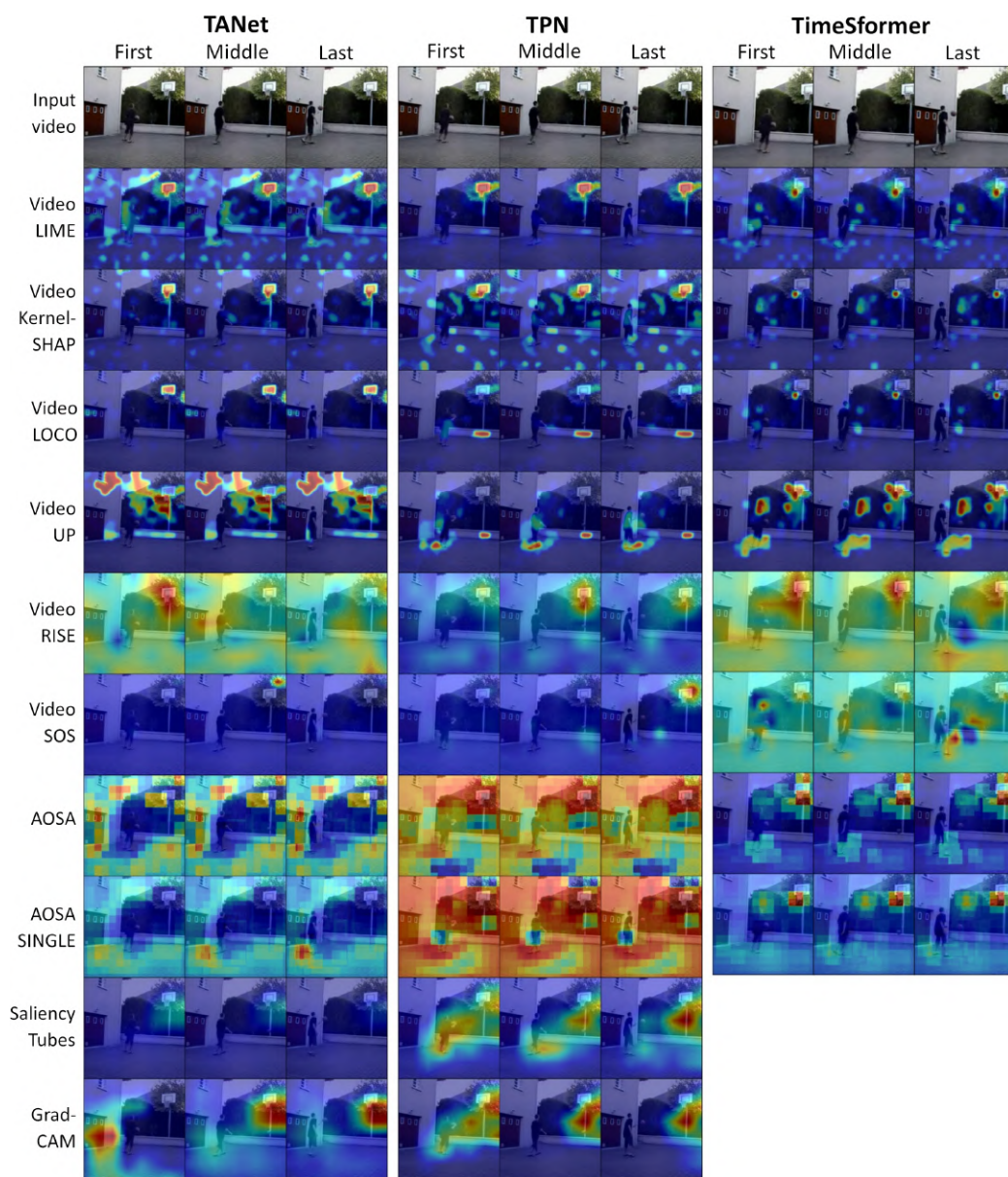


Figure A.14: Explanations for the UCF101 class “basketball.”

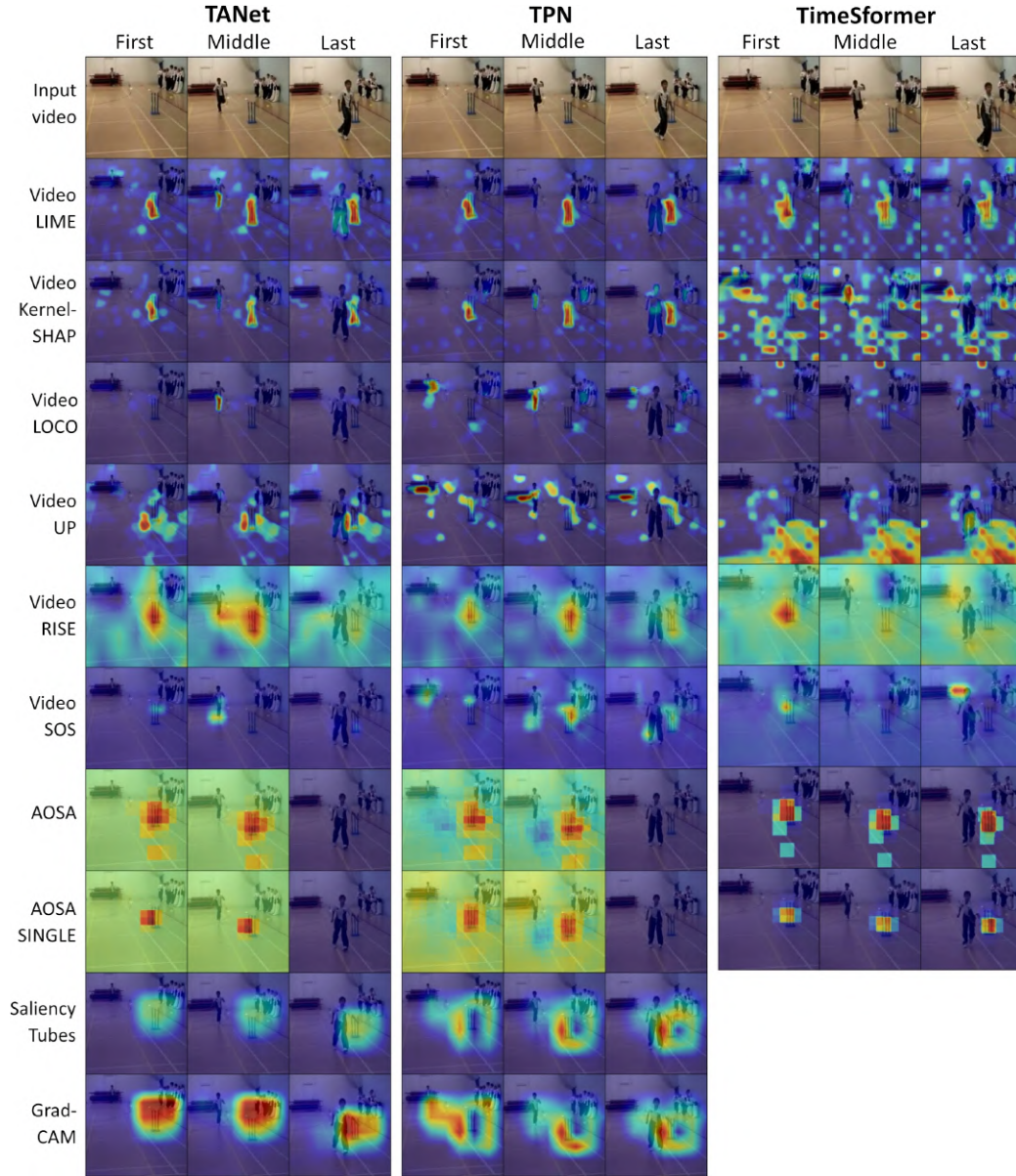


Figure A.15: Explanations for the UCF101 class “cricket bowling.”

Appendix B. Results by dataset and model

This section provides further insight into the deletion and insertion metrics. See Figure B.16 and Figure B.17. For both metrics, results are grouped

by network, dataset, and method.

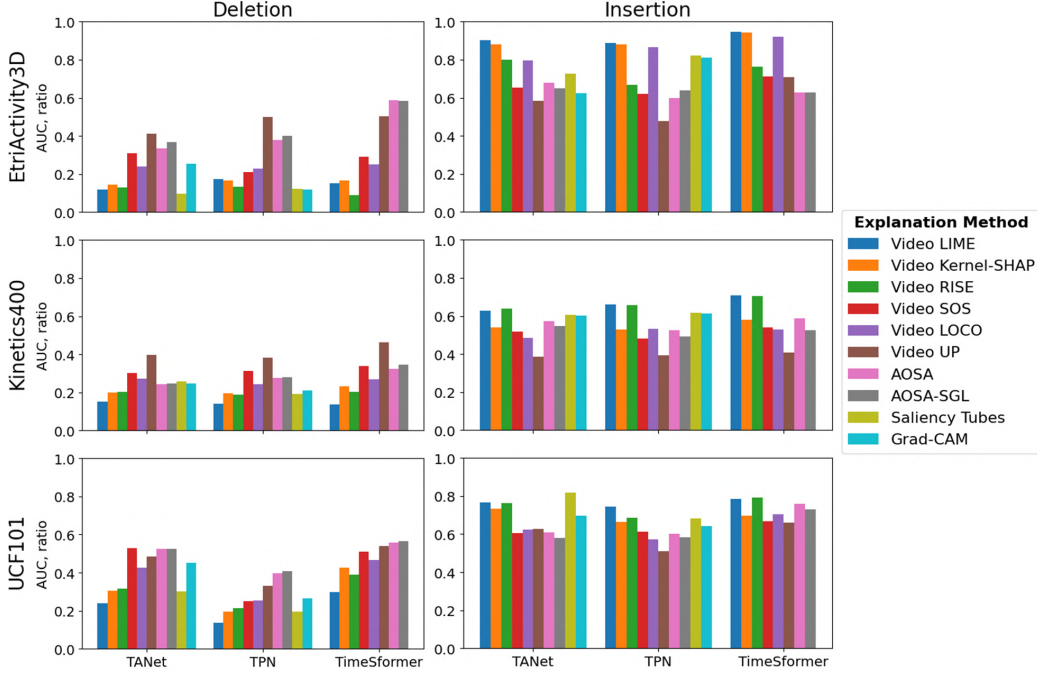


Figure B.16: AUC (ratio) by method, network, and dataset. In the left column, deletion game (lower values are better); in the right one, insertion game (higher values are better). By rows, the datasets (ETRI-Activity3D, Kinetics 400, and UCF101) and within each plot, the networks (TANet, TPN, and TimeSformer) and explanation methods (see legend).

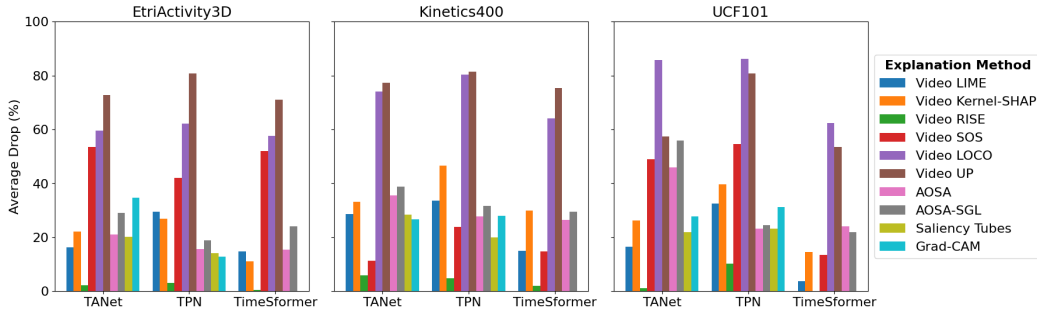


Figure B.17: Average drop (%) by dataset, network, and method (lower is better). By columns, the datasets, ETRI-Activity3D, Kinetics 400, and UCF101, and within each plot, the networks (TimeSformer, TANet and TPN) and explanation methods (see legend).

CRedit authorship contribution statement

F. Xavier Gaya-Morey: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing Preparation, Visualization, Project administration. **Jose M. Buades-Rubio:** Conceptualization, Methodology, Formal Analysis, Writing - Review & Editing Preparation, Supervision, Project administration, Funding acquisition. **I. Scott MacKenzie:** Writing - Review & Editing Preparation, Supervision. **Cristina Manresa-Yee:** Conceptualization, Methodology, Writing - Review & Editing Preparation, Supervision, Project administration, Funding acquisition.

Funding sources

This work is part of the Project PID2023-149079OB-I00 (EXPLAINME) funded by MICIU/AEI/10.13039/ 501100011033/ and FEDER, EU and of Project PID2022-136779OB-C32 (PLEISAR) funded by MICIU/AEI/10.13039/ 501100011033/ and FEDER, EU. F. X. Gaya-Morey was supported by an FPU scholarship from the Ministry of European Funds, University and Culture of the Government of the Balearic Islands.

Data availability

The REVEX framework is made publicly available at https://github.com/Xavi3398/revex_framework.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins,

- R. Chatila, F. Herrera, Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI, *Information Fusion* 58 (2020) 82–115. doi:10.1016/j.inffus.2019.12.012.
- [2] K. Simonyan, A. Vedaldi, A. Zisserman, Deep inside convolutional networks: Visualising image classification models and saliency maps (2014). arXiv:1312.6034, doi:10.48550/arXiv.1312.6034.
- [3] S. Bach, A. Binder, G. Montavon, F. Klauschen, K.-R. Müller, W. Samek, On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation, *PLOS ONE* 10 (7) (2015) 1–46. doi:10.1371/journal.pone.0130140.
- [4] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: 2017 IEEE International Conference on Computer Vision (ICCV), IEEE, New York, 2017, pp. 618–626. doi:10.1109/ICCV.2017.74.
- [5] T. Hartley, K. Sidorov, C. Willis, D. Marshall, SWAG-V: Explanations for video using superpixels weighted by average gradients, in: 2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), IEEE, New York, 2022, pp. 1576–1585. doi:10.1109/WACV51458.2022.00164.
- [6] A. Stergiou, G. Kapidis, G. Kalliatakis, C. Chrysoulas, R. Veltkamp, R. Poppe, Saliency tubes: Visual explanations for spatio-temporal convolutions, in: 2019 IEEE International Conference on Image Processing (ICIP), IEEE, New York, 2019, pp. 1830–1834. doi:10.1109/ICIP.2019.8803153.
- [7] L. Hiley, A. Preece, Y. Hicks, S. Chakraborty, P. Gurram, R. Tomsett, Explaining motion relevance for activity recognition in video deep learning models (2020). arXiv:2003.14285, doi:10.48550/arXiv.2003.14285.
- [8] M. T. Ribeiro, S. Singh, C. Guestrin, “Why should I trust you?”: Explaining the predictions of any classifier, in: Proceedings of the

- 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16, ACM, New York, 2016, p. 1135–1144. doi:10.1145/2939672.2939778.
- [9] V. Petsiuk, A. Das, K. Saenko, Rise: Randomized input sampling for explanation of black-box models (2018). [arXiv:1806.07421](#), doi:10.48550/arXiv.1806.07421.
 - [10] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17, Curran Associates Inc., Red Hook, NY, USA, 2017, p. 4768–4777.
 - [11] F. X. Gaya-Morey, C. Manresa-Yee, J. M. Buades-Rubio, Explainable activity recognition for the elderly, in: Proceedings of the XXIII International Conference on Human Computer Interaction, Interacción '23, ACM, New York, 2024. doi:10.1145/3612783.3612790.
 - [12] T. Uchiyama, N. Sogi, K. Niinuma, K. Fukui, Visually explaining 3D-CNN predictions for video classification with an adaptive occlusion sensitivity analysis, in: 2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), IEEE, New York, 2023, pp. 1513–1522. doi:10.1109/WACV56688.2023.00156.
 - [13] Z. Li, W. Wang, Z. Li, Y. Huang, Y. Sato, Towards visually explaining video understanding networks with perturbation, in: 2021 IEEE Winter Conference on Applications of Computer Vision (WACV), IEEE, New York, 2021, pp. 1119–1128. doi:10.1109/WACV48630.2021.00116.
 - [14] A. Stergiou, G. Kapidis, G. Kalliatakis, C. Chrysoulas, R. Poppe, R. Veltkamp, Class feature pyramids for video explanation, in: 2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW), IEEE, New York, 2019, pp. 4255–4264. doi:10.1109/ICCVW.2019.00524.
 - [15] L. Hiley, A. Preece, Y. Hicks, D. Marshall, H. Taylor, Discriminating spatial and temporal relevance in deep Taylor decompositions for explainable activity recognition (2019). [arXiv:1908.01536](#), doi:10.48550/arXiv.1908.01536.

- [16] S. A. Bargal, A. Zunino, D. Kim, J. Zhang, V. Murino, S. Sclaroff, Excitation backprop for RNNs, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, IEEE, New York, 2018, pp. 1440–1449. doi:10.1109/CVPR.2018.00156.
- [17] C. Anders, G. Montavon, W. Samek, K.-R. Müller, Understanding patch-based learning by explaining predictions (2018). arXiv:1806.06926, doi:10.48550/arXiv.1806.06926.
- [18] G. Montavon, S. Lapuschkin, A. Binder, W. Samek, K.-R. Müller, Explaining nonlinear classification decisions with deep Taylor decomposition, *Pattern Recognition* 65 (2017) 211–222. doi:10.1016/j.patcog.2016.11.008.
- [19] V. Srinivasan, S. Lapuschkin, C. Hellge, K.-R. Müller, W. Samek, Interpretable human action recognition in compressed domain, in: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, New York, 2017, pp. 1692–1696. doi:10.1109/ICASSP.2017.7952445.
- [20] C. Gan, N. Wang, Y. Yang, D.-Y. Yeung, A. G. Hauptmann, Devnet: A deep event network for multimedia event detection and evidence recounting, in: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, New York, 2015, pp. 2568–2577. doi:10.1109/CVPR.2015.7298872.
- [21] M. D. Zeiler, R. Fergus, Visualizing and understanding convolutional networks, in: D. Fleet, T. Pajdla, B. Schiele, T. Tuytelaars (Eds.), *Computer Vision – ECCV 2014*, Springer International Publishing, Cham, 2014, pp. 818–833. doi:10.1007/978-3-319-10590-1_53.
- [22] T. Hartley, K. Sidorov, C. Willis, D. Marshall, SWAG: superpixels weighted by average gradients for explanations of CNNs, in: 2021 IEEE Winter Conference on Applications of Computer Vision (WACV), IEEE, New York, 2021, pp. 423–432. doi:10.1109/WACV48630.2021.00047.
- [23] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, S. Süsstrunk, SLIC superpixels, Technical Report, EPFL (2010) 15.

- [24] J. T. Springenberg, A. Dosovitskiy, T. Brox, M. Riedmiller, Striving for simplicity: The all convolutional net (2015). [arXiv:1412.6806](#), [doi:10.48550/arXiv.1412.6806](#).
- [25] R. Fong, M. Patrick, A. Vedaldi, Understanding deep networks via extremal perturbations and smooth masks, in: 2019 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE, New York, 2019, pp. 2950–2958. [doi:10.1109/ICCV.2019.00304](#).
- [26] I. C. Covert, S. Lundberg, S.-I. Lee, Explaining by removing: A unified framework for model explanation, *Journal of Machine Learning Research* 22 (1) (2021) 1–90, <http://jmlr.org/papers/v22/20-1316.html>.
- [27] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, *Journal of the American Statistical Association* 113 (523) (2018) 1094–1111. [doi:10.1080/01621459.2017.1307116](#).
- [28] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, *Journal of Machine Learning Research* 3 (null) (2003) 1157–1182, <https://www.jmlr.org/papers/v3/guyon03a.html>.
- [29] D. Sun, X. Yang, M.-Y. Liu, J. Kautz, PWC-Net: CNNs for optical flow using pyramid, warping, and cost volume, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, IEEE, New York, 2018, pp. 8934–8943. [doi:10.1109/CVPR.2018.00931](#).
- [30] P. F. Felzenszwalb, D. P. Huttenlocher, Efficient graph-based image segmentation, *International Journal of Computer Vision* 59 (2) (2004) 167–181. [doi:10.1023/B:VISI.0000022288.19776.77](#).
- [31] A. Vedaldi, S. Soatto, Quick shift and kernel methods for mode seeking, in: D. Forsyth, P. Torr, A. Zisserman (Eds.), *Computer Vision – ECCV 2008*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 705–718. [doi:10.1007/978-3-540-88693-8_52](#).
- [32] P. Neubert, P. Protzel, Compact watershed and preemptive SLIC: on improving trade-offs of superpixel segmentation algorithms, in: 2014 22nd International Conference on Pattern Recognition, 2014, pp. 996–1001. [doi:10.1109/ICPR.2014.181](#).

- [33] M. Miró-Nicolau, A. J. i Capó, G. Moyà-Alcover, Assessing fidelity in XAI post-hoc techniques: A comparative study with ground truth explanations datasets, *Artificial Intelligence* 335 (2024) 104179. doi: 10.1016/j.artint.2024.104179.
- [34] R. C. Fong, A. Vedaldi, Interpretable explanations of black boxes by meaningful perturbation, in: 2017 IEEE International Conference on Computer Vision (ICCV), IEEE, New York, 2017, pp. 3449–3457. doi: 10.1109/ICCV.2017.371.
- [35] J. Xie, L. Xu, E. Chen, Image denoising and inpainting with deep neural networks, in: *Proceedings of the 25th International Conference on Neural Information Processing Systems - Volume 1, NIPS’12*, Curran Associates Inc., Red Hook, NY, 2012, p. 341–349.
- [36] A. Hudon, T. Demazure, A. Karran, P.-M. Léger, S. Sénécal, Explainable artificial intelligence (xai): How the visualization of ai predictions affects user cognitive load and confidence, in: F. D. Davis, R. Riedl, J. vom Brocke, P.-M. Léger, A. B. Randolph, G. Müller-Putz (Eds.), *Information Systems and Neuroscience*, Springer International Publishing, Cham, 2021, pp. 237–246. doi:10.1007/978-3-030-88900-5_27.
- [37] G. Bertasius, H. Wang, L. Torresani, Is space-time attention all you need for video understanding? (2021). arXiv:2102.05095, doi:10.48550/arXiv.2102.05095.
- [38] Z. Liu, L. Wang, W. Wu, C. Qian, T. Lu, TAM: temporal adaptive module for video recognition, in: 2021 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE, New York, 2021, pp. 13688–13698. doi:10.1109/ICCV48922.2021.01345.
- [39] C. Yang, Y. Xu, J. Shi, B. Dai, B. Zhou, Temporal pyramid network for action recognition, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, New York, 2020, pp. 588–597. doi:10.1109/CVPR42600.2020.00067.
- [40] MMAAction2 Contributors, Openmmlab’s next generation video understanding toolbox and benchmark (2020).
URL <https://github.com/open-mmlab/mmaaction2>

- [41] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, in: International Conference on Learning Representations, 2021.
URL <https://openreview.net/forum?id=YicbFdNTTy>
- [42] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, F. Viola, T. Green, T. Back, P. Natsev, M. Suleyman, A. Zisserman, The kinetics human action video dataset (2017). [arXiv:1705.06950](#), doi:10.48550/arXiv.1705.06950.
- [43] K. Soomro, A. R. Zamir, M. Shah, Ucf101: A dataset of 101 human actions classes from videos in the wild (2012). [arXiv:1212.0402](#), doi:10.48550/arXiv.1212.0402.
- [44] J. Jang, D. Kim, C. Park, M. Jang, J. Lee, J. Kim, ETRI-Activity3D: a large-scale rgb-d dataset for robots to recognize daily activities of the elderly, in: 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, New York, 2020, pp. 10990–10997. doi:10.1109/IROS45743.2020.9341160.
- [45] Y.-G. Jiang, J. Liu, A. Roshan Zamir, I. Laptev, M. Piccardi, M. Shah, R. Sukthankar, Thumos challenge: Action recognition with a large number of classes (2013).
URL <https://www.crcv.ucf.edu/ICCV13-Action-Workshop>
- [46] V. Petsiuk, R. Jain, V. Manjunatha, V. I. Morariu, A. Mehra, V. Ordonez, K. Saenko, Black-box explanation of object detectors via saliency maps, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, New York, 2021, pp. 11438–11447. doi:10.1109/CVPR46437.2021.01128.
- [47] A. Chattopadhyay, A. Sarkar, P. Howlader, V. N. Balasubramanian, Grad-CAM++: generalized gradient-based visual explanations for deep convolutional networks, in: 2018 IEEE Winter Conference on Applications of Computer Vision (WACV), IEEE, New York, 2018, pp. 839–847. doi:10.1109/WACV.2018.00097.

- [48] C. Cao, X. Liu, Y. Yang, Y. Yu, J. Wang, Z. Wang, Y. Huang, L. Wang, C. Huang, W. Xu, D. Ramanan, T. S. Huang, Look and think twice: Capturing top-down visual attention with feedback convolutional neural networks, in: 2015 IEEE International Conference on Computer Vision (ICCV), IEEE, New York, 2015, pp. 2956–2964. doi:10.1109/ICCV.2015.338.
- [49] J. Zhang, S. A. Bargal, Z. Lin, J. Brandt, X. Shen, S. Sclaroff, Top-down neural attention by excitation backprop, International Journal of Computer Vision 126 (10) (2018) 1084–1102. doi:10.1007/s11263-017-1059-x.